

Chapter 5

The Structure and Function of Large Biological Molecules

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings

Overview: The Molecules of Life

- All living things are made up of four classes of large biological molecules: carbohydrates, lipids, proteins, and nucleic acids
- Within cells, small organic molecules are joined together to form larger molecules
- **Macromolecules** are large molecules composed of thousands of covalently connected atoms
- Molecular structure and function are inseparable

Fig. 5-1



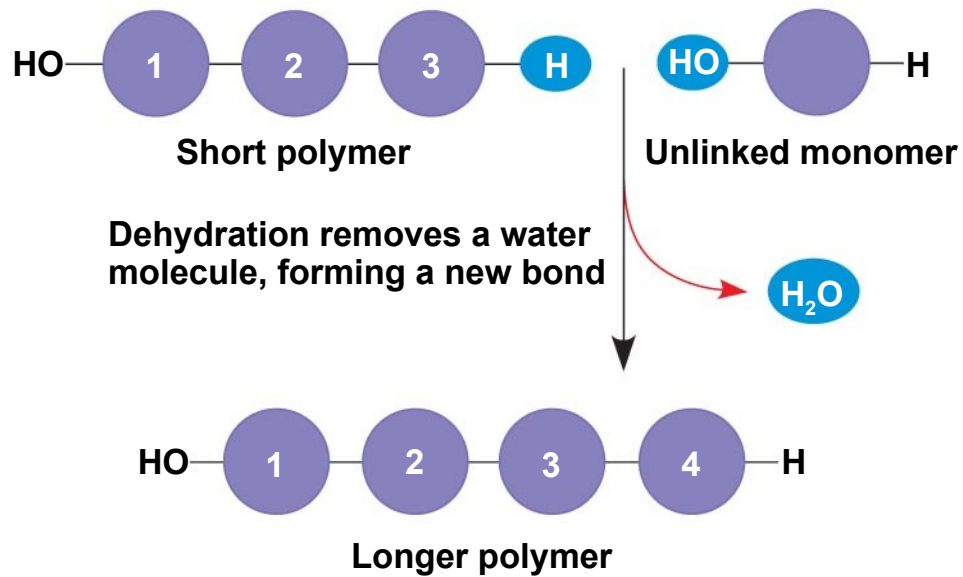
Concept 5.1: Macromolecules are polymers, built from monomers

- A **polymer** is a long molecule consisting of many similar building blocks
- These small building-block molecules are called **monomers**
- Three of the four classes of life's organic molecules are polymers:
 - Carbohydrates
 - Proteins
 - Nucleic acids

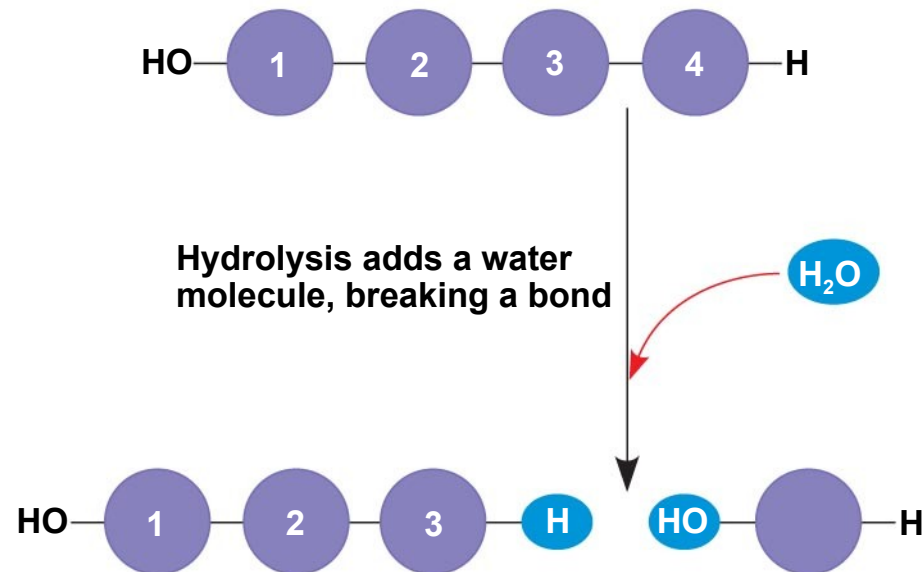
The Synthesis and Breakdown of Polymers

- A **condensation reaction** or more specifically a **dehydration reaction** occurs when two monomers bond together through the loss of a water molecule
- **Enzymes** are macromolecules that speed up the dehydration process
- Polymers are disassembled to monomers by **hydrolysis**, a reaction that is essentially the reverse of the dehydration reaction

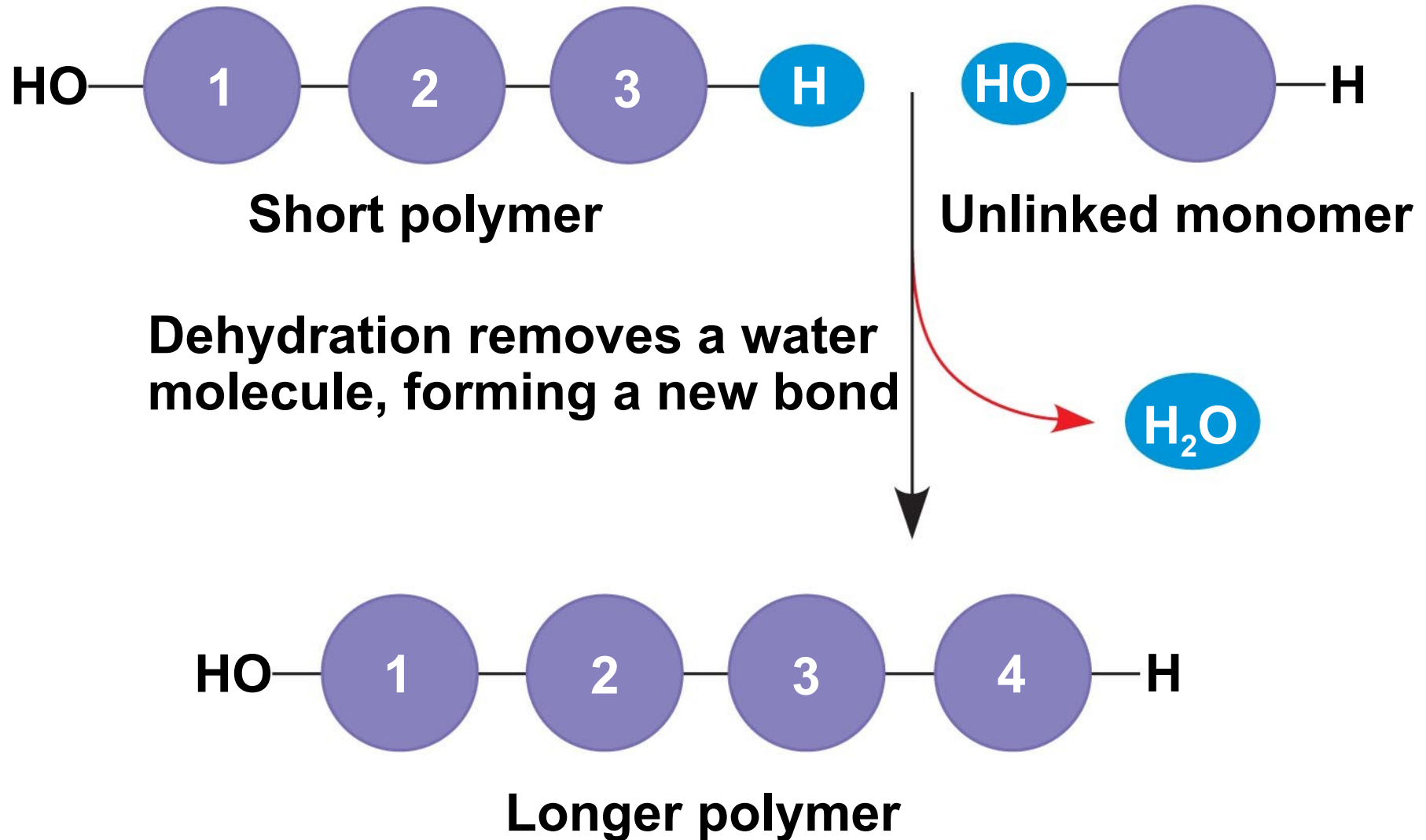
Fig. 5-2



(a) Dehydration reaction in the synthesis of a polymer

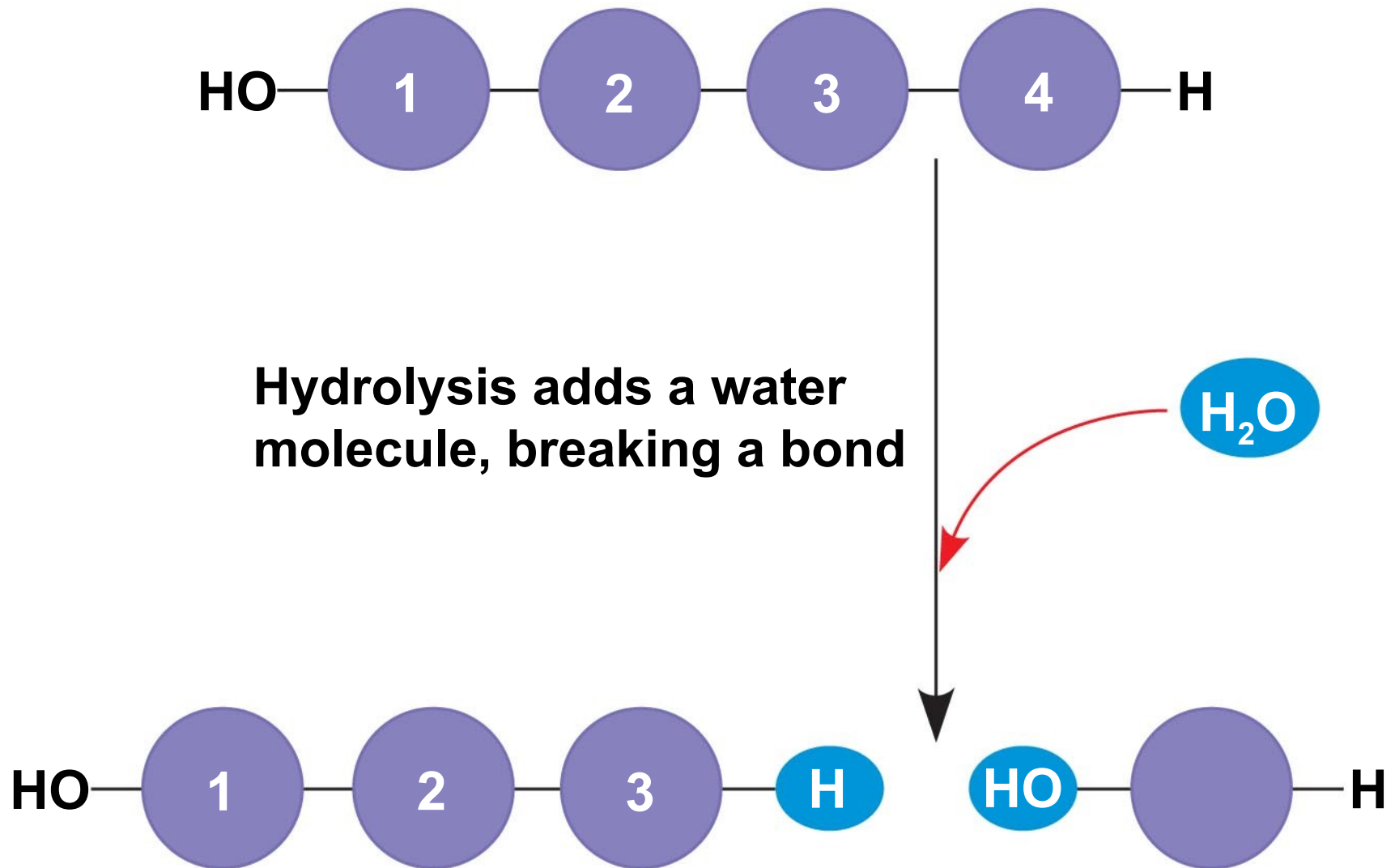


(b) Hydrolysis of a polymer



(a) Dehydration reaction in the synthesis of a polymer

Fig. 5-2b



(b) Hydrolysis of a polymer

The Diversity of Polymers

- Each cell has thousands of different kinds of macromolecules
- Macromolecules vary among cells of an organism, vary more within a species, and vary even more between species
- An immense variety of polymers can be built from a small set of monomers

Concept 5.2: Carbohydrates serve as fuel and building material

- **Carbohydrates** include sugars and the polymers of sugars
- The simplest carbohydrates are monosaccharides, or single sugars
- Carbohydrate macromolecules are polysaccharides, polymers composed of many sugar building blocks

Sugars

- **Monosaccharides** have molecular formulas that are usually multiples of CH_2O
- Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is the most common monosaccharide
- Monosaccharides are classified by
 - The location of the carbonyl group (as aldose or ketose)
 - The number of carbons in the carbon skeleton

Fig. 5-3

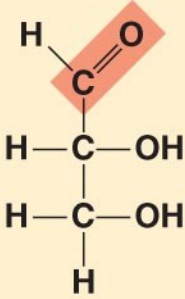
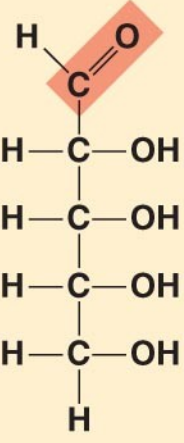
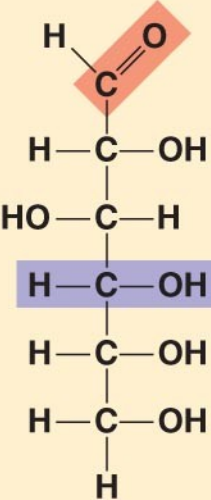
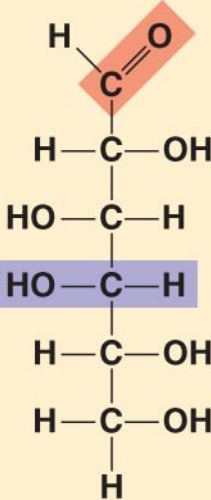
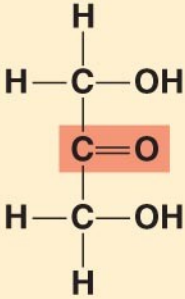
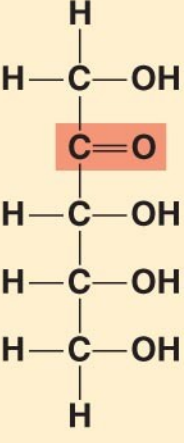
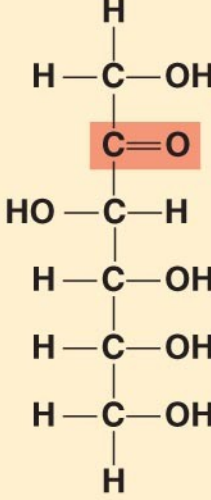
	Trioses ($C_3H_6O_3$)	Pentoses ($C_5H_{10}O_5$)	Hexoses ($C_6H_{12}O_6$)	
Aldoses	 Glyceraldehyde	 Ribose	 Glucose	 Galactose
Ketoses	 Dihydroxyacetone	 Ribulose	 Fructose	

Fig. 5-3a

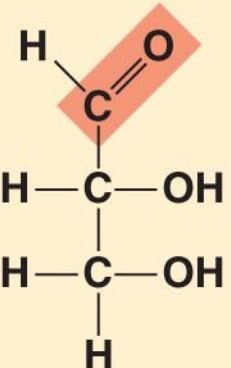
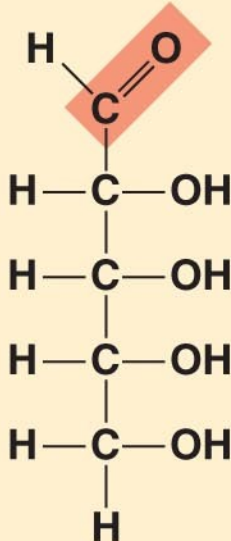
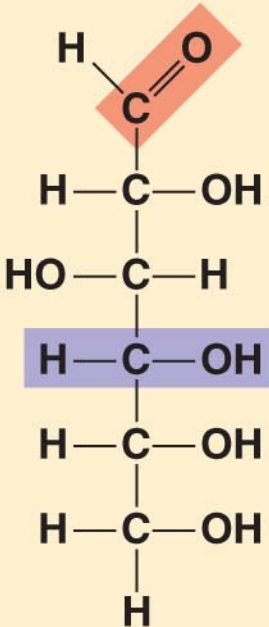
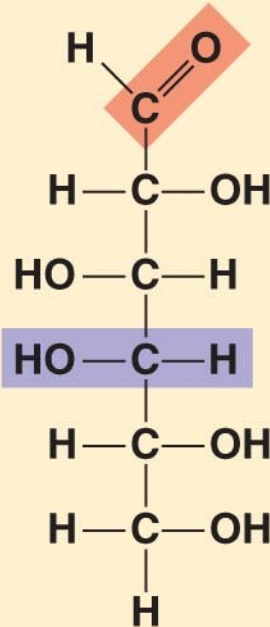
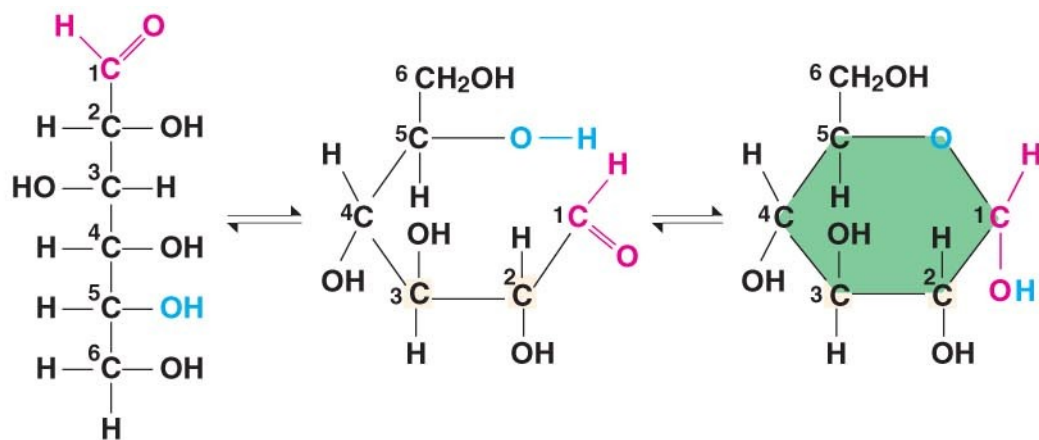
	Trioses ($C_3H_6O_3$)	Pentoses ($C_5H_{10}O_5$)	Hexoses ($C_6H_{12}O_6$)	
Aldoses	 Glyceraldehyde	 Ribose	 Glucose	 Galactose

Fig. 5-3b

	Trioses ($C_3H_6O_3$)	Pentoses ($C_5H_{10}O_5$)	Hexoses ($C_6H_{12}O_6$)
Ketoses	$ \begin{array}{c} H \\ \\ H-C-OH \\ \\ C=O \\ \\ H-C-OH \\ \\ H \end{array} $ Dihydroxyacetone	$ \begin{array}{c} H \\ \\ H-C-OH \\ \\ C=O \\ \\ H-C-OH \\ \\ H-C-OH \\ \\ H-C-OH \\ \\ H \end{array} $ Ribulose	$ \begin{array}{c} H \\ \\ H-C-OH \\ \\ C=O \\ \\ HO-C-H \\ \\ H-C-OH \\ \\ H-C-OH \\ \\ H-C-OH \\ \\ H \end{array} $ Fructose

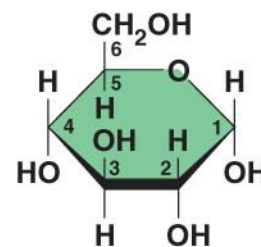
-
- Though often drawn as linear skeletons, in aqueous solutions many sugars form rings
 - Monosaccharides serve as a major fuel for cells and as raw material for building molecules

Fig. 5-4

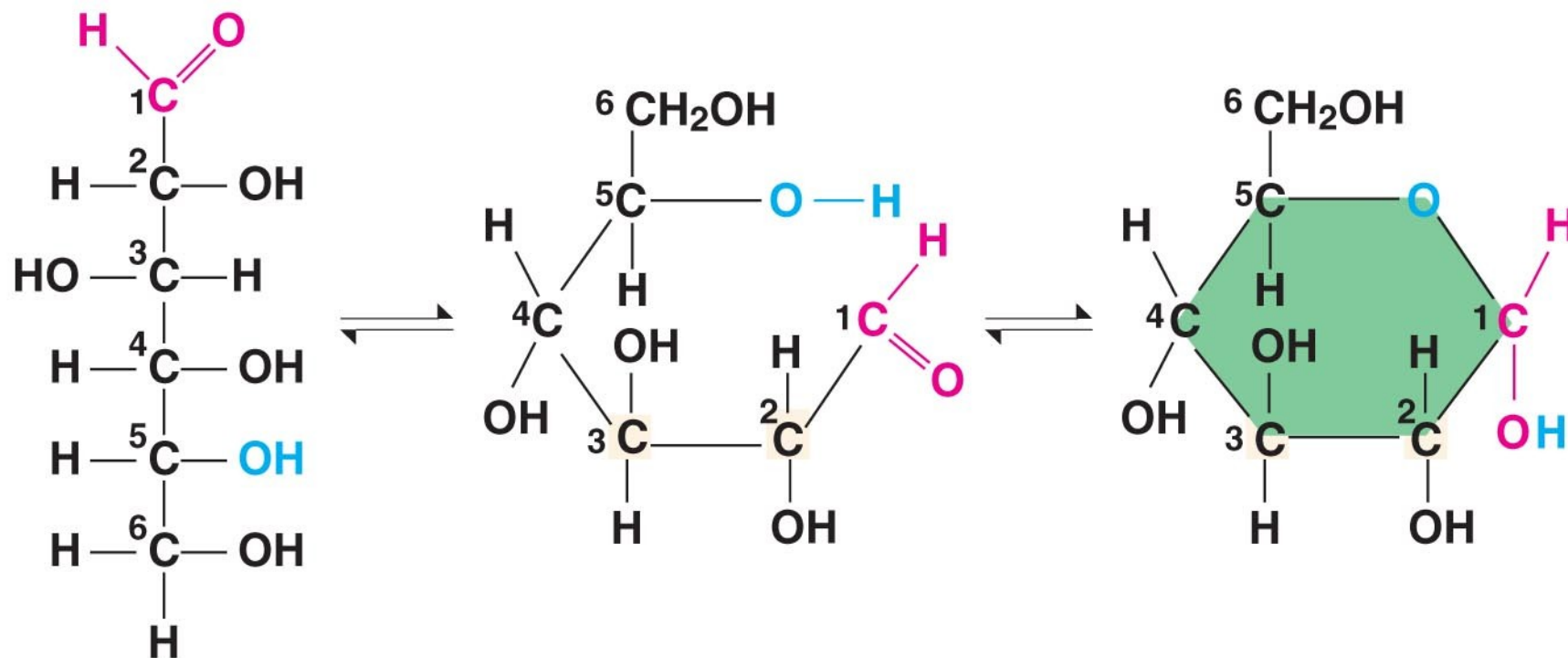


(a) Linear and ring forms

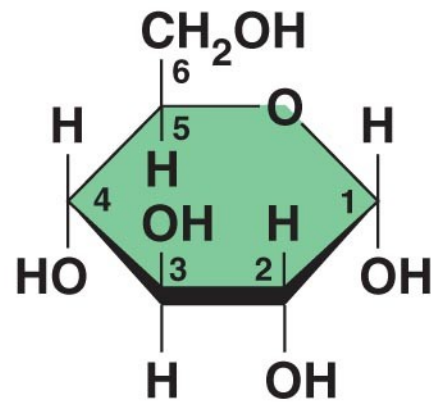
Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.



(b) Abbreviated ring structure



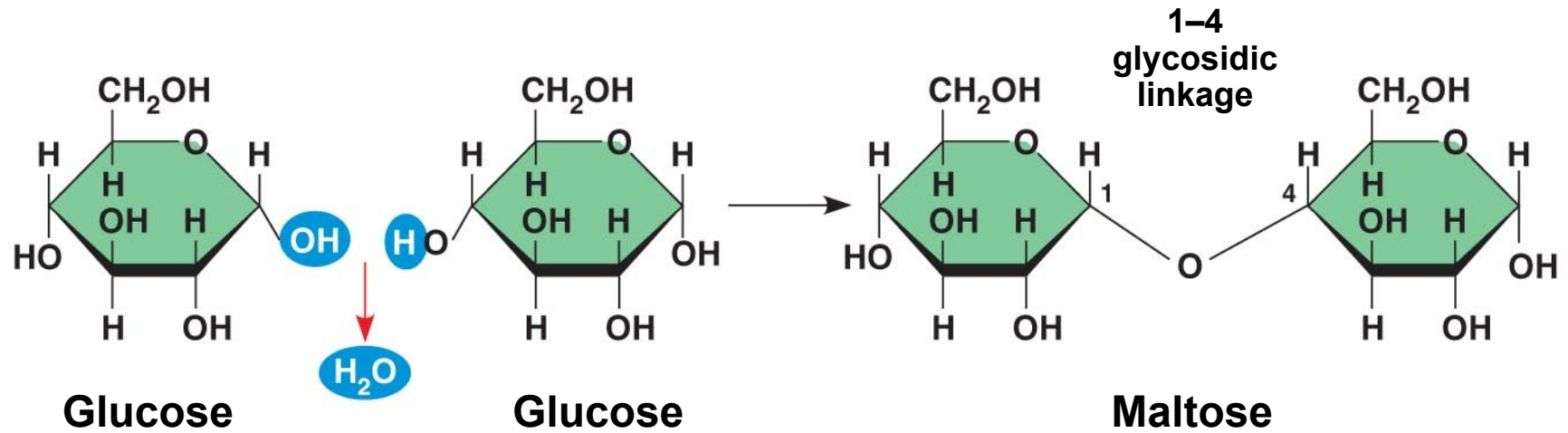
(a) Linear and ring forms



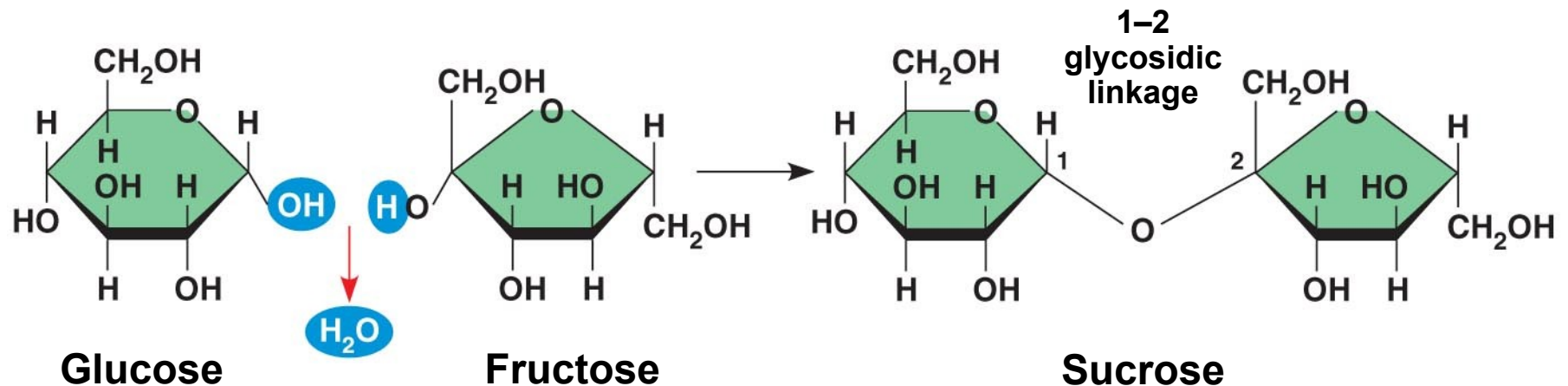
(b) Abbreviated ring structure

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

-
- A **disaccharide** is formed when a dehydration reaction joins two monosaccharides
 - This covalent bond is called a **glycosidic linkage**



(a) Dehydration reaction in the synthesis of maltose



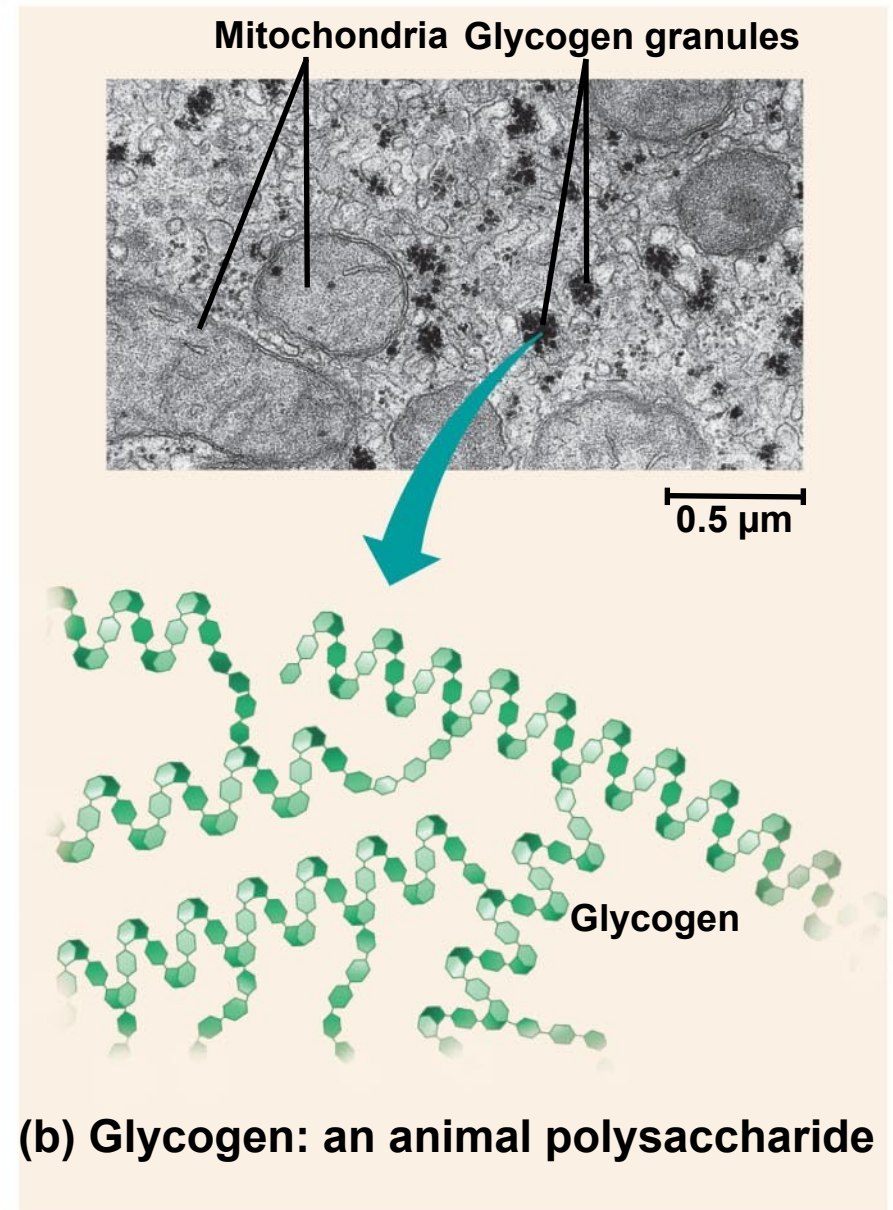
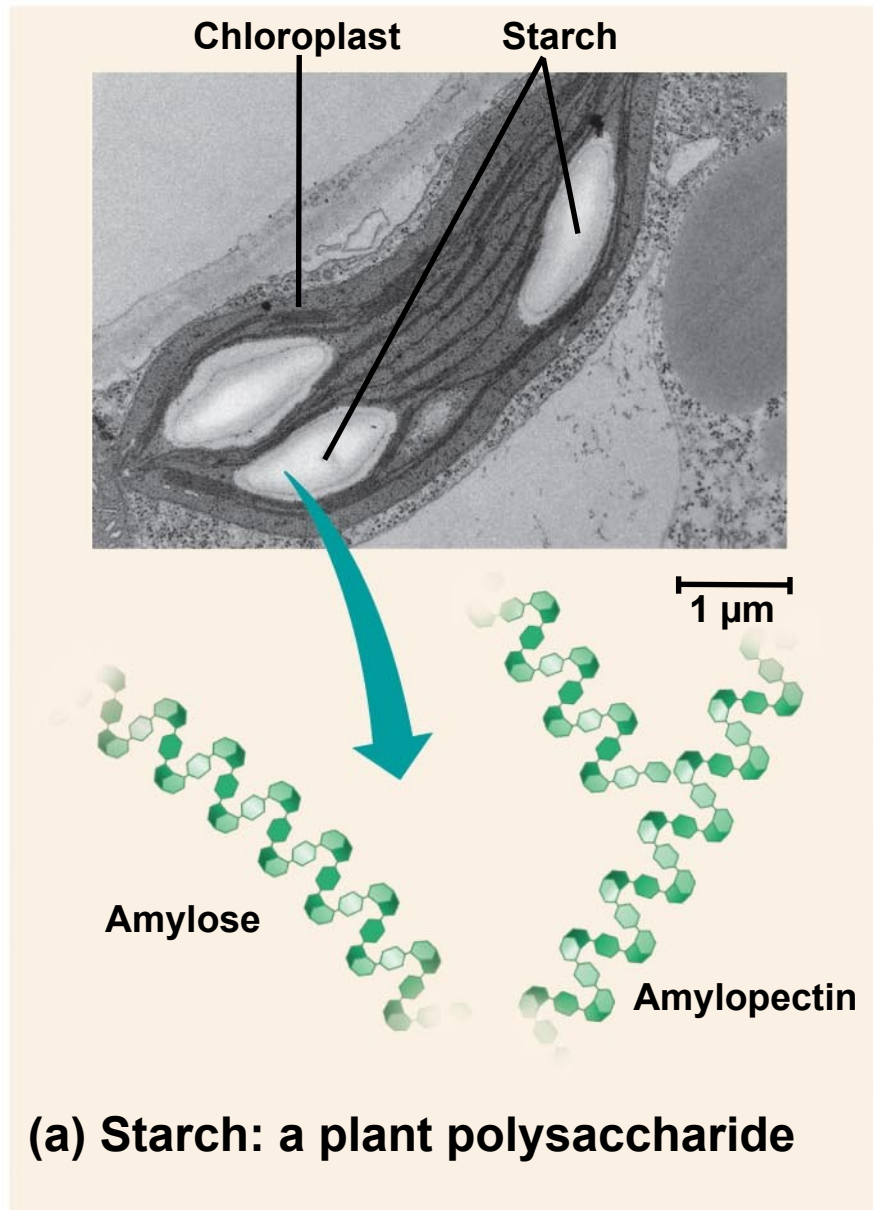
(b) Dehydration reaction in the synthesis of sucrose

Polysaccharides

- **Polysaccharides**, the polymers of sugars, have storage and structural roles
- The structure and function of a polysaccharide are determined by its sugar monomers and the positions of glycosidic linkages

Storage Polysaccharides

- **Starch**, a storage polysaccharide of plants, consists entirely of glucose monomers
- Plants store surplus starch as granules within chloroplasts and other plastids



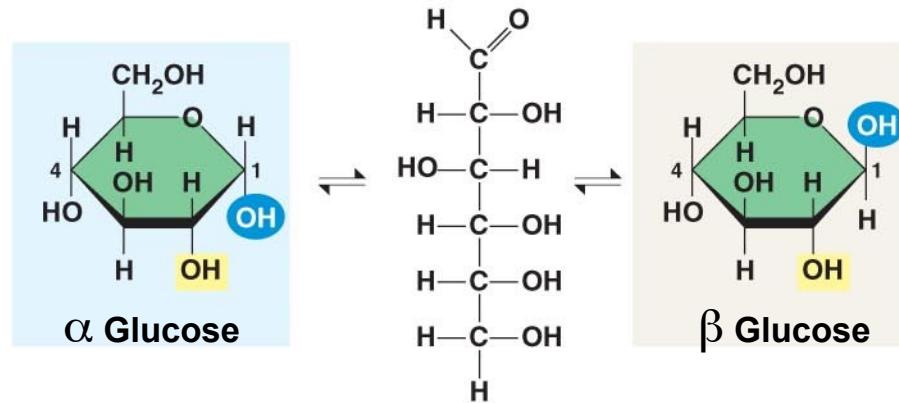
-
- **Glycogen** is a storage polysaccharide in animals
 - Humans and other vertebrates store glycogen mainly in liver and muscle cells

Structural Polysaccharides

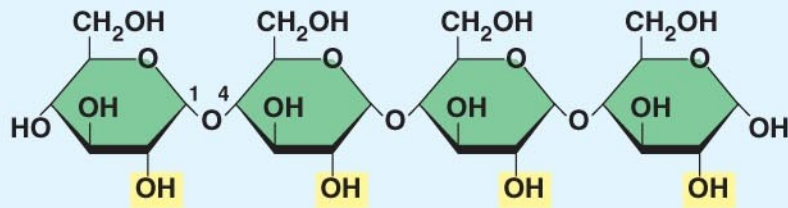
- The polysaccharide **cellulose** is a major component of the tough wall of plant cells
- Like starch, cellulose is a polymer of glucose, but the glycosidic linkages differ
- The difference is based on two ring forms for glucose: alpha (α) and beta (β)□

Fig. 5-7

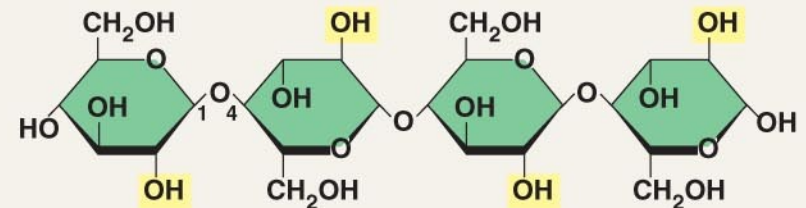
(a) α and β glucose ring structures

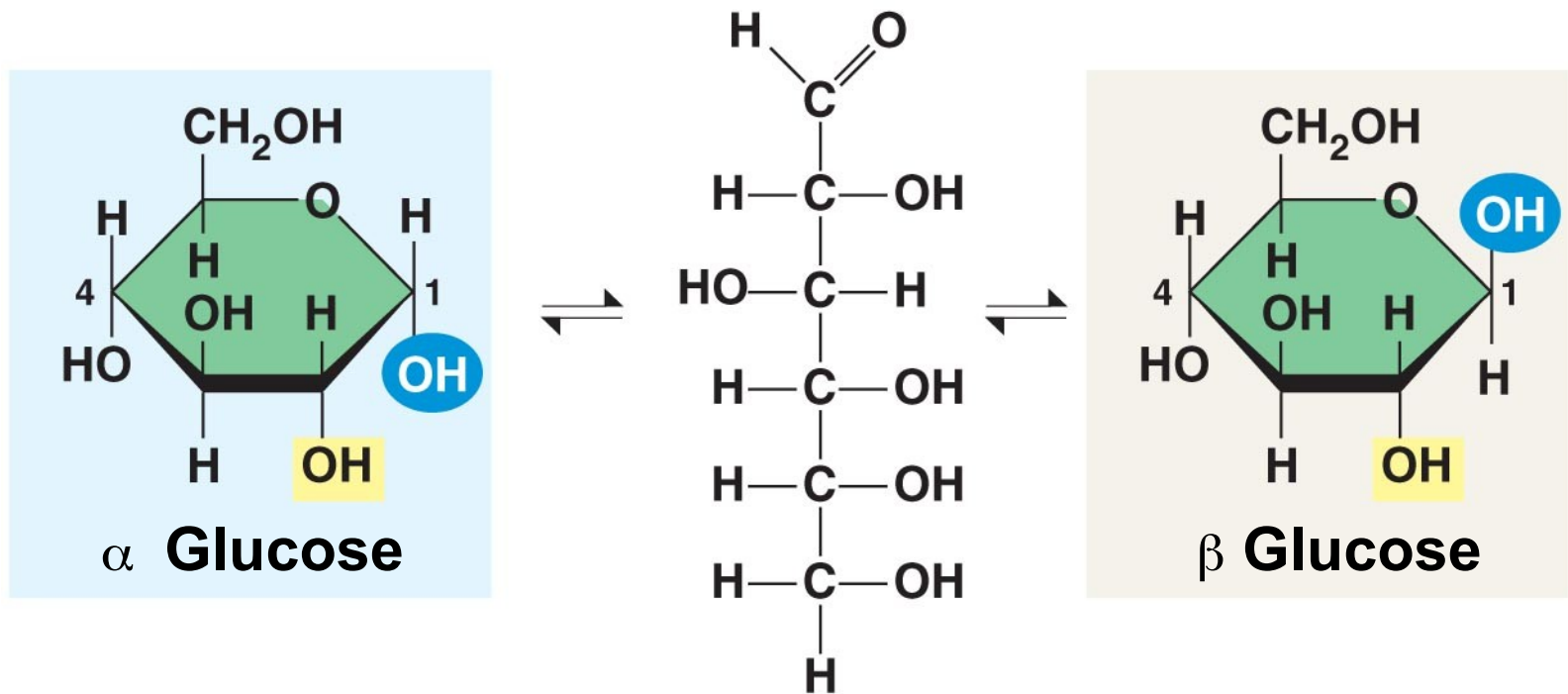


(b) Starch: 1–4 linkage of α glucose monomers



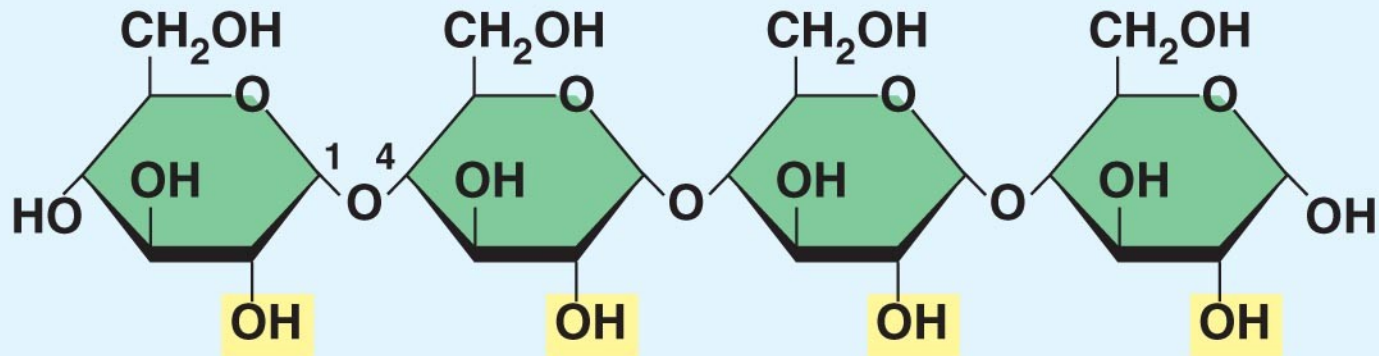
(b) Cellulose: 1–4 linkage of β glucose monomers



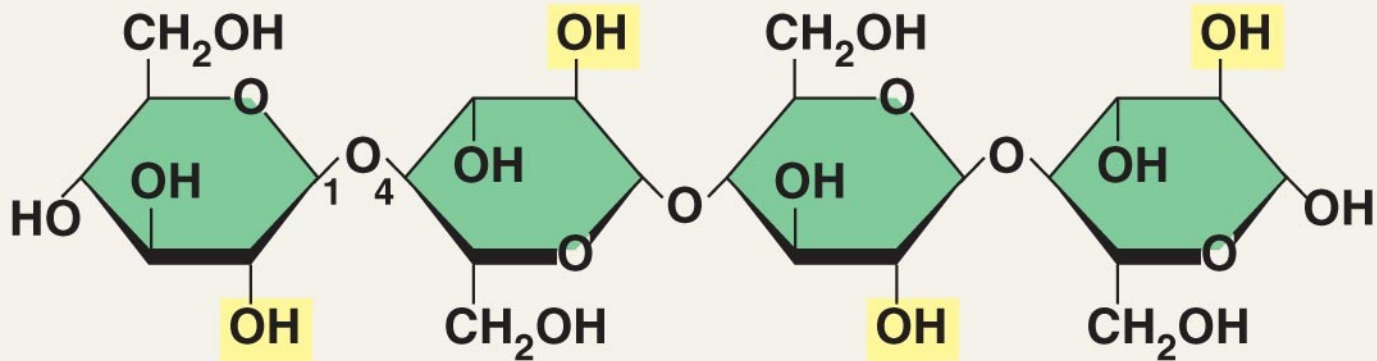


(a) α and β glucose ring structures

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.



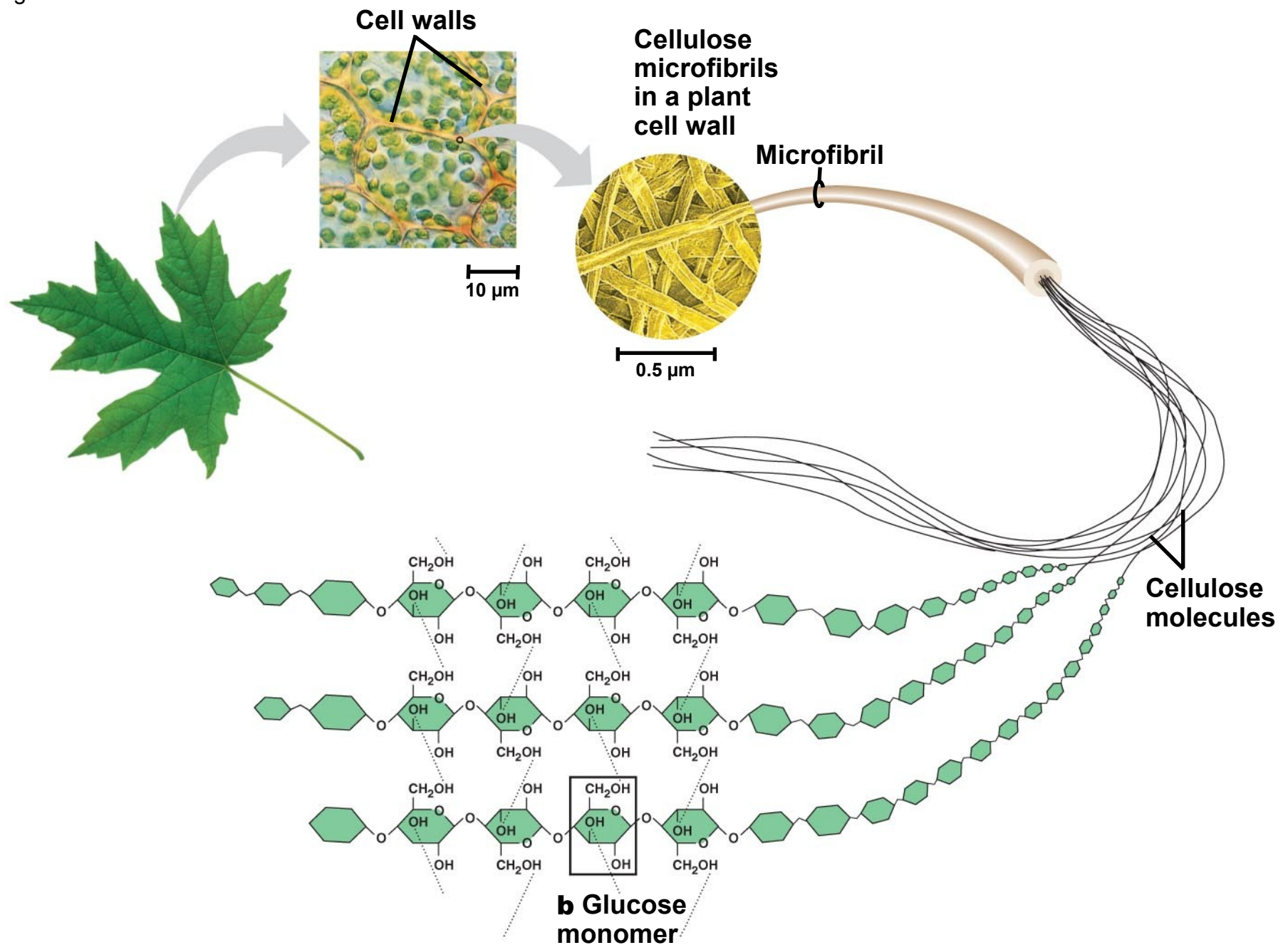
(b) Starch: 1–4 linkage of α glucose monomers



(c) Cellulose: 1–4 linkage of β glucose monomers

-
- Polymers with α glucose are helical
 - Polymers with β glucose are straight
 - In straight structures, H atoms on one strand can bond with OH groups on other strands
 - Parallel cellulose molecules held together this way are grouped into microfibrils, which form strong building materials for plants

Fig. 5-8



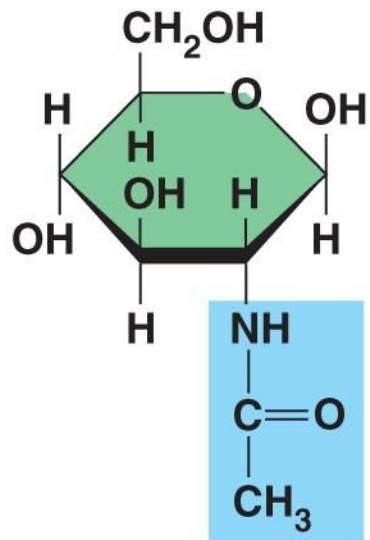
-
- Enzymes that digest starch by hydrolyzing α linkages can't hydrolyze β linkages in cellulose
 - Cellulose in human food passes through the digestive tract as insoluble fiber
 - Some microbes use enzymes to digest cellulose
 - Many herbivores, from cows to termites, have symbiotic relationships with these microbes

Fig. 5-9



Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

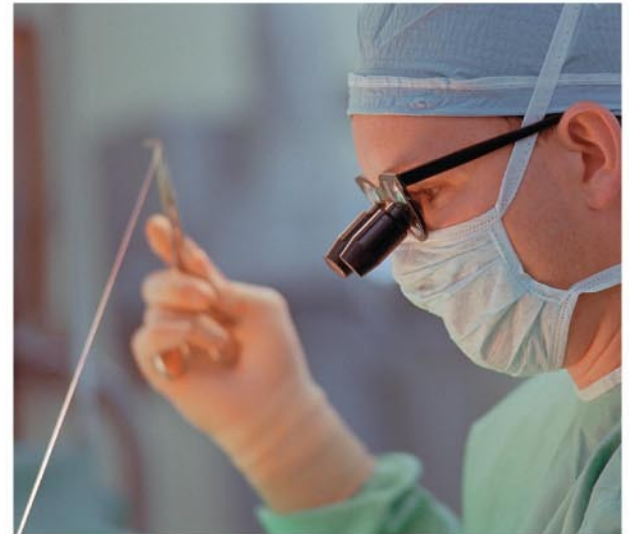
-
- **Chitin**, another structural polysaccharide, is found in the exoskeleton of arthropods
 - Chitin also provides structural support for the cell walls of many fungi



(a) The structure of the chitin monomer.



(b) Chitin forms the exoskeleton of arthropods.



(c) Chitin is used to make a strong and flexible surgical thread.

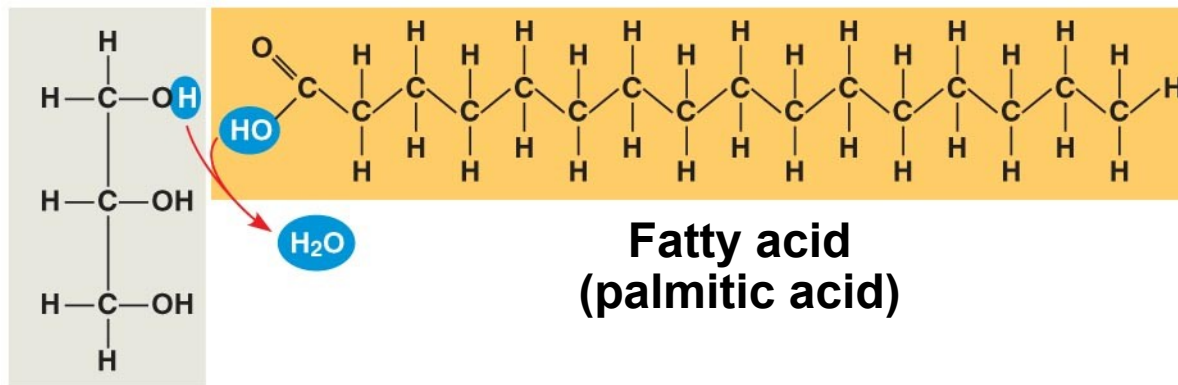
Concept 5.3: Lipids are a diverse group of hydrophobic molecules

- **Lipids** are the one class of large biological molecules that do not form polymers
- The unifying feature of lipids is having little or no affinity for water
- Lipids are hydrophobic because they consist mostly of hydrocarbons, which form nonpolar covalent bonds
- The most biologically important lipids are fats, phospholipids, and steroids

Fats

- **Fats** are constructed from two types of smaller molecules: glycerol and fatty acids
- Glycerol is a three-carbon alcohol with a hydroxyl group attached to each carbon
- A **fatty acid** consists of a carboxyl group attached to a long carbon skeleton

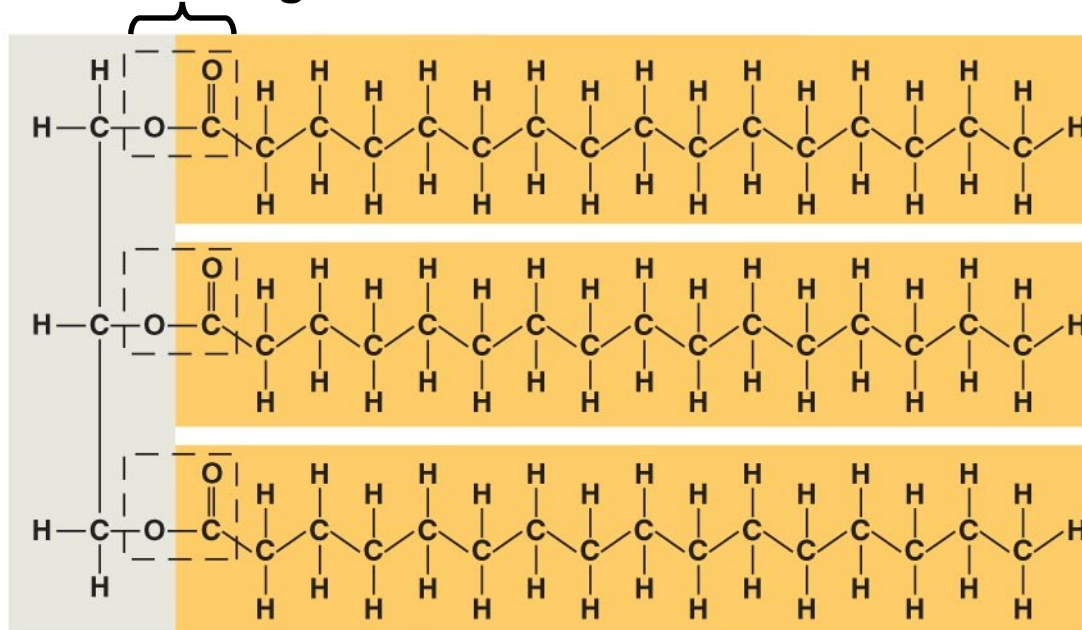
Fig. 5-11



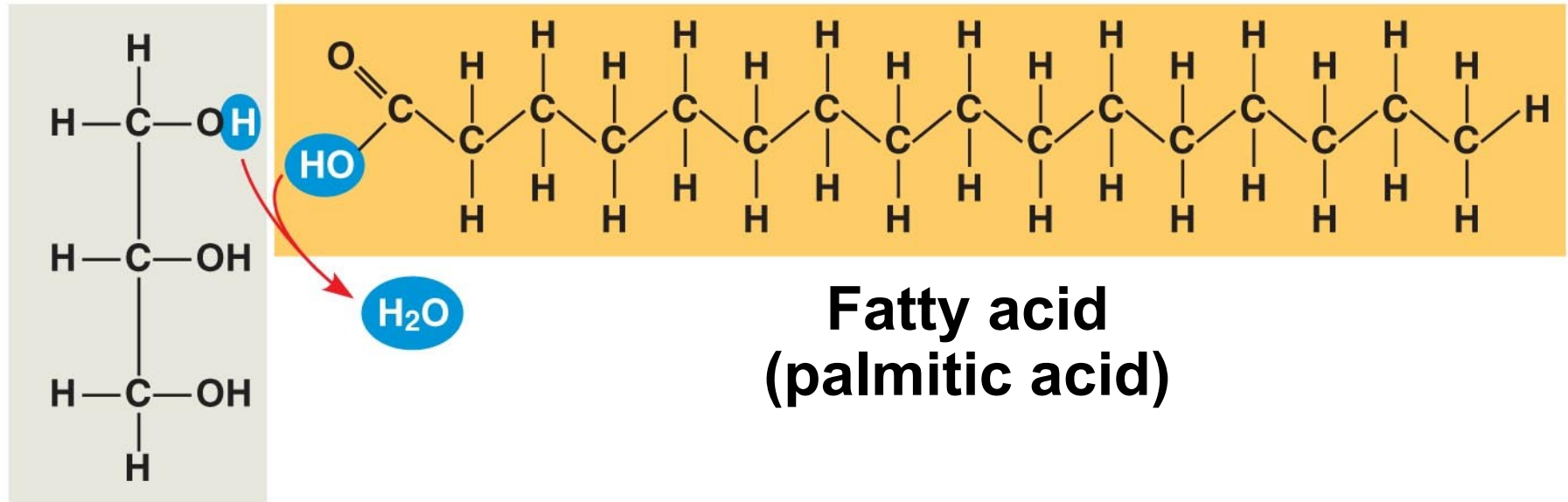
Glycerol

(a) Dehydration reaction in the synthesis of a fat

Ester linkage



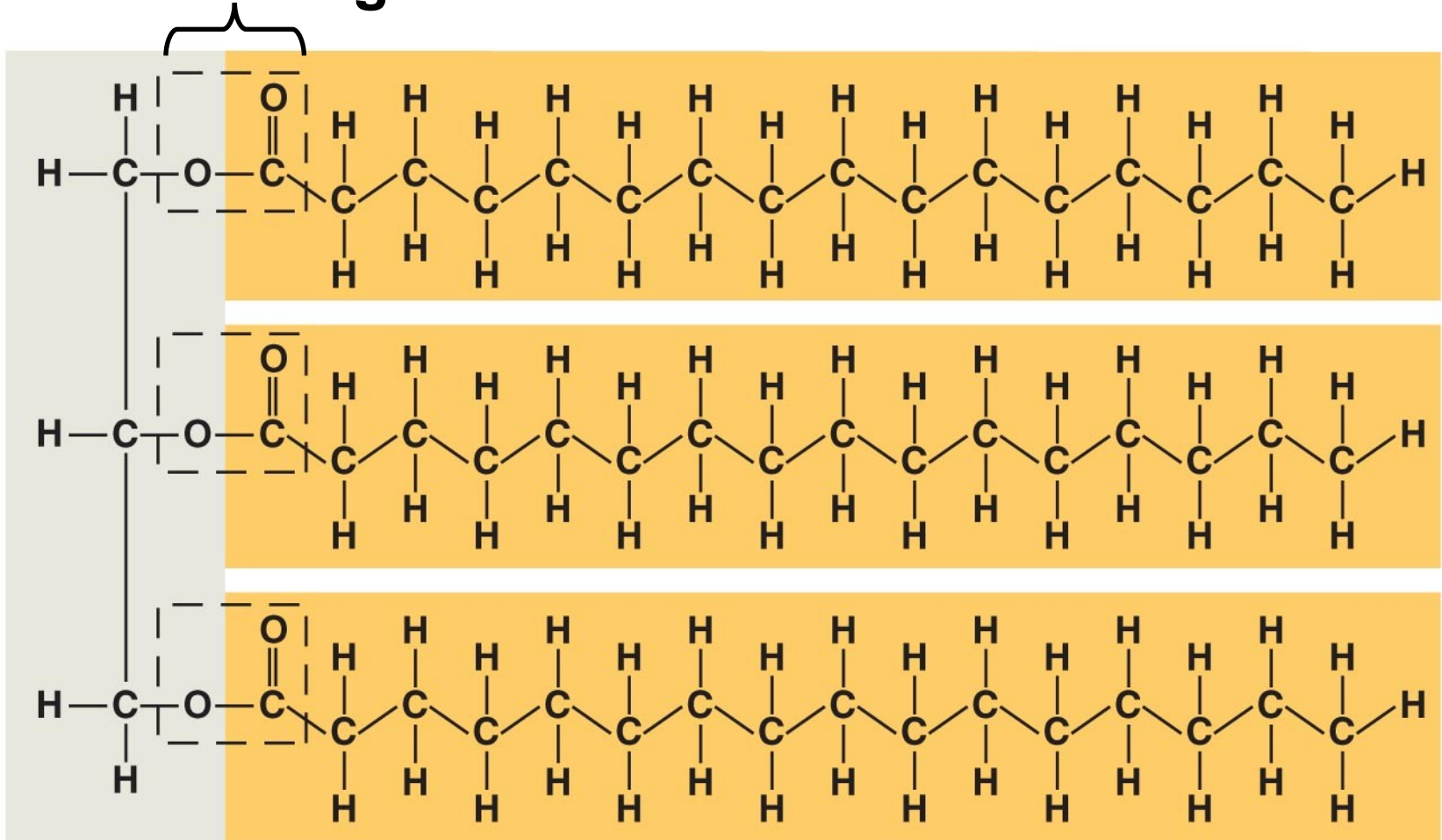
(b) Fat molecule (triacylglycerol)



Glycerol

(a) Dehydration reaction in the synthesis of a fat

Ester linkage



(b) Fat molecule (triacylglycerol)

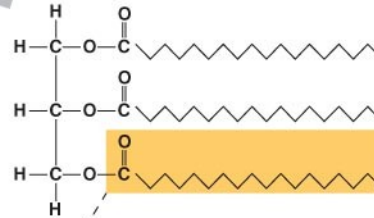
-
- Fats separate from water because water molecules form hydrogen bonds with each other and exclude the fats
 - In a fat, three fatty acids are joined to glycerol by an ester linkage, creating a **triacylglycerol**, or triglyceride

-
- Fatty acids vary in length (number of carbons) and in the number and locations of double bonds
 - **Saturated fatty acids** have the maximum number of hydrogen atoms possible and no double bonds
 - **Unsaturated fatty acids** have one or more double bonds

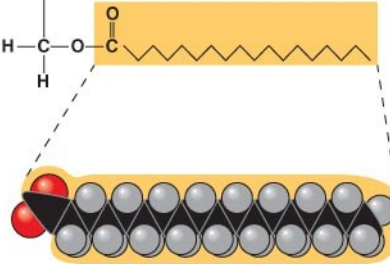
Fig. 5-12



Structural formula of a saturated fat molecule



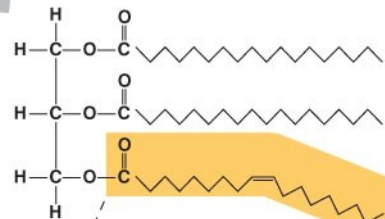
Stearic acid, a saturated fatty acid



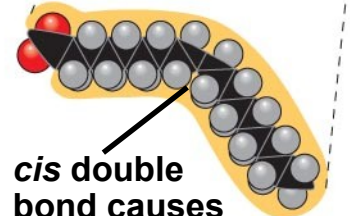
(a) Saturated fat



Structural formula of an unsaturated fat molecule



Oleic acid, an unsaturated fatty acid

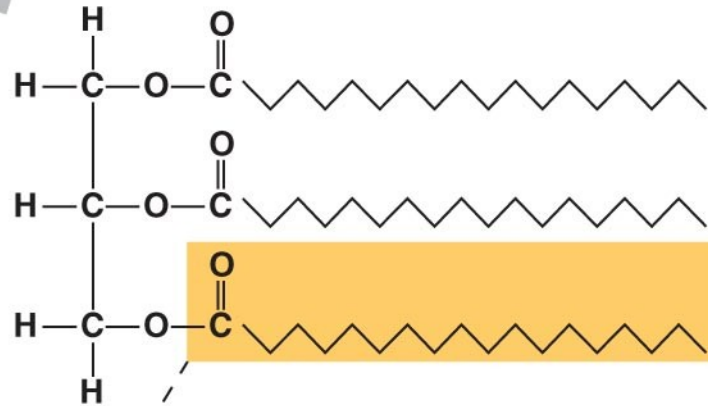


***cis* double bond causes bending**

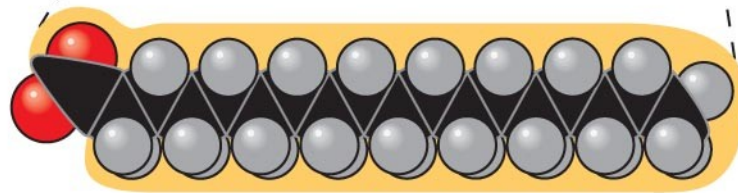
(b) Unsaturated fat



**Structural
formula of a
saturated fat
molecule**

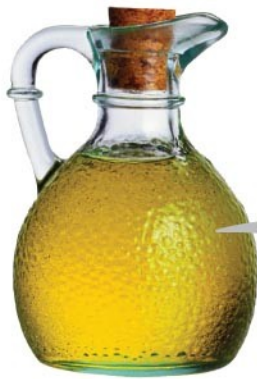


**Stearic acid, a
saturated fatty
acid**

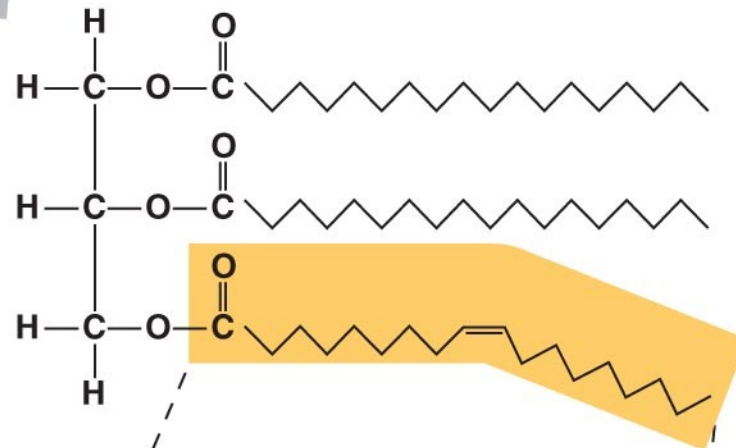


(a) Saturated fat

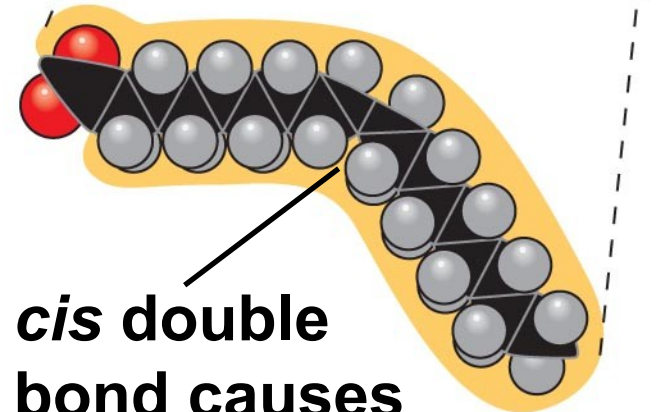
Fig. 5-12b



**Structural formula
of an unsaturated
fat molecule**



**Oleic acid, an
unsaturated
fatty acid**



***cis* double
bond causes
bending**

(b) Unsaturated fat

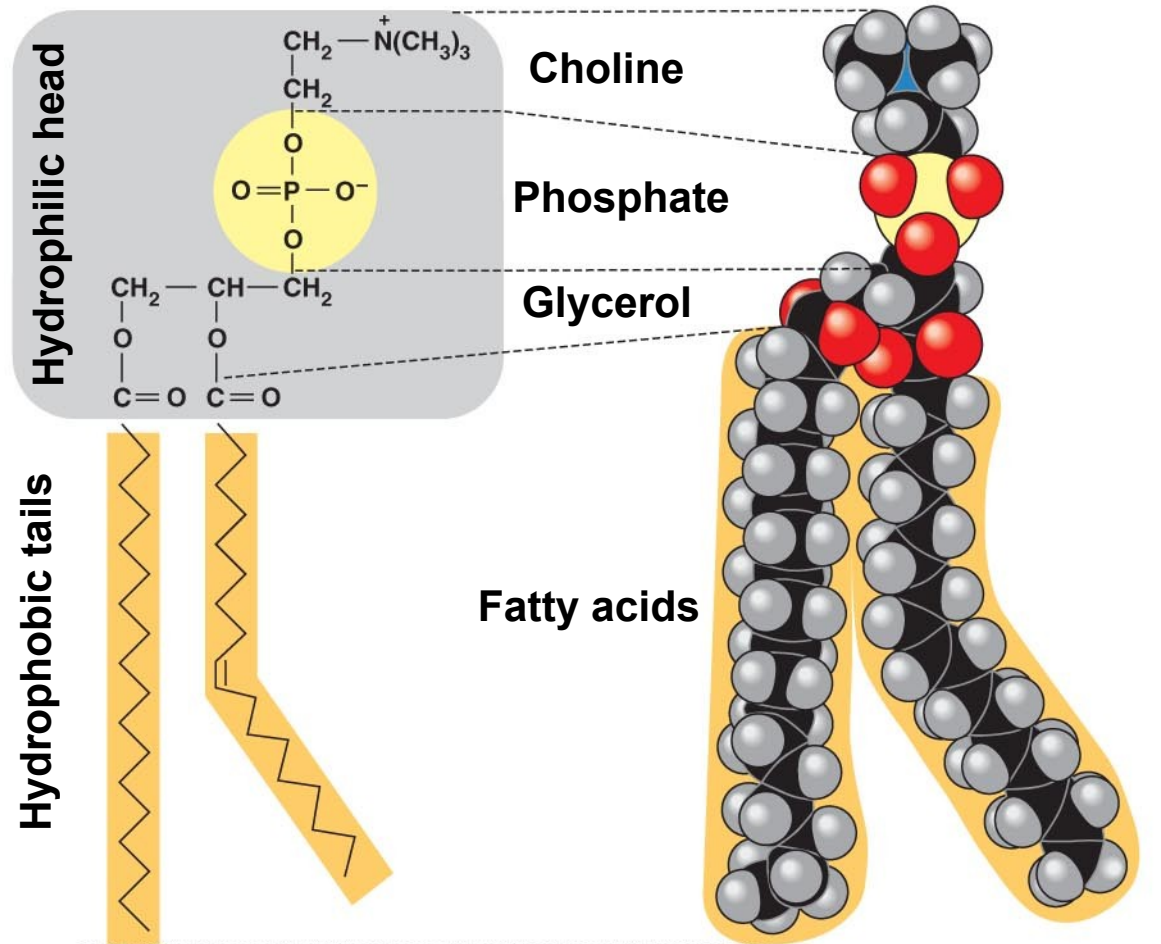
-
- Fats made from saturated fatty acids are called saturated fats, and are solid at room temperature
 - Most animal fats are saturated
 - Fats made from unsaturated fatty acids are called unsaturated fats or oils, and are liquid at room temperature
 - Plant fats and fish fats are usually unsaturated

-
- A diet rich in saturated fats may contribute to cardiovascular disease through plaque deposits
 - Hydrogenation is the process of converting unsaturated fats to saturated fats by adding hydrogen
 - Hydrogenating vegetable oils also creates unsaturated fats with *trans* double bonds
 - These *trans* fats may contribute more than saturated fats to cardiovascular disease

-
- The major function of fats is energy storage
 - Humans and other mammals store their fat in adipose cells
 - Adipose tissue also cushions vital organs and insulates the body

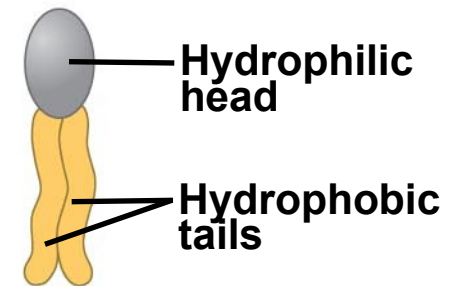
Phospholipids

- In a **phospholipid**, two fatty acids and a phosphate group are attached to glycerol
- The two fatty acid tails are hydrophobic, but the phosphate group and its attachments form a hydrophilic head



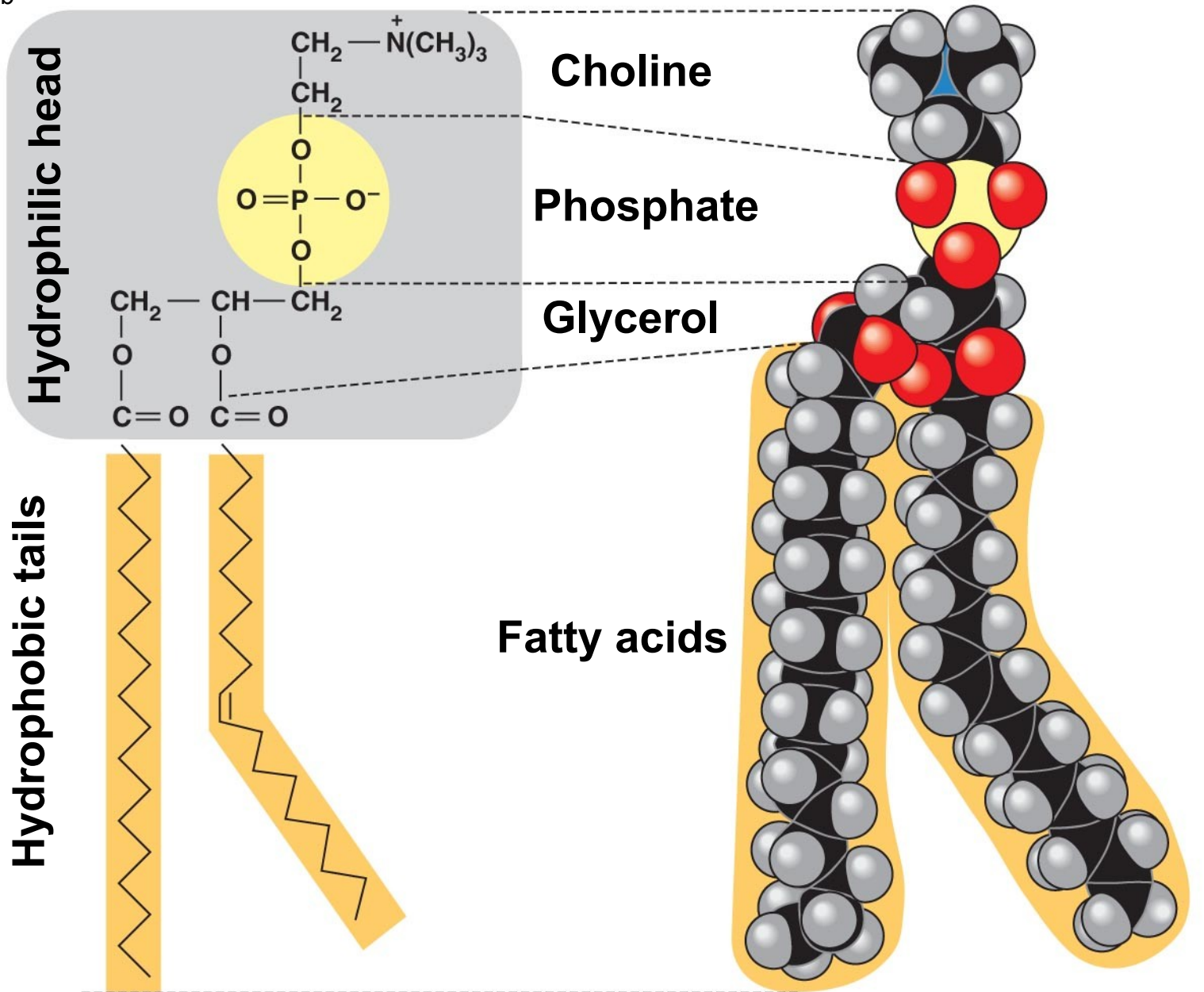
(a) Structural formula

(b) Space-filling model



(c) Phospholipid symbol

Fig. 5-13ab

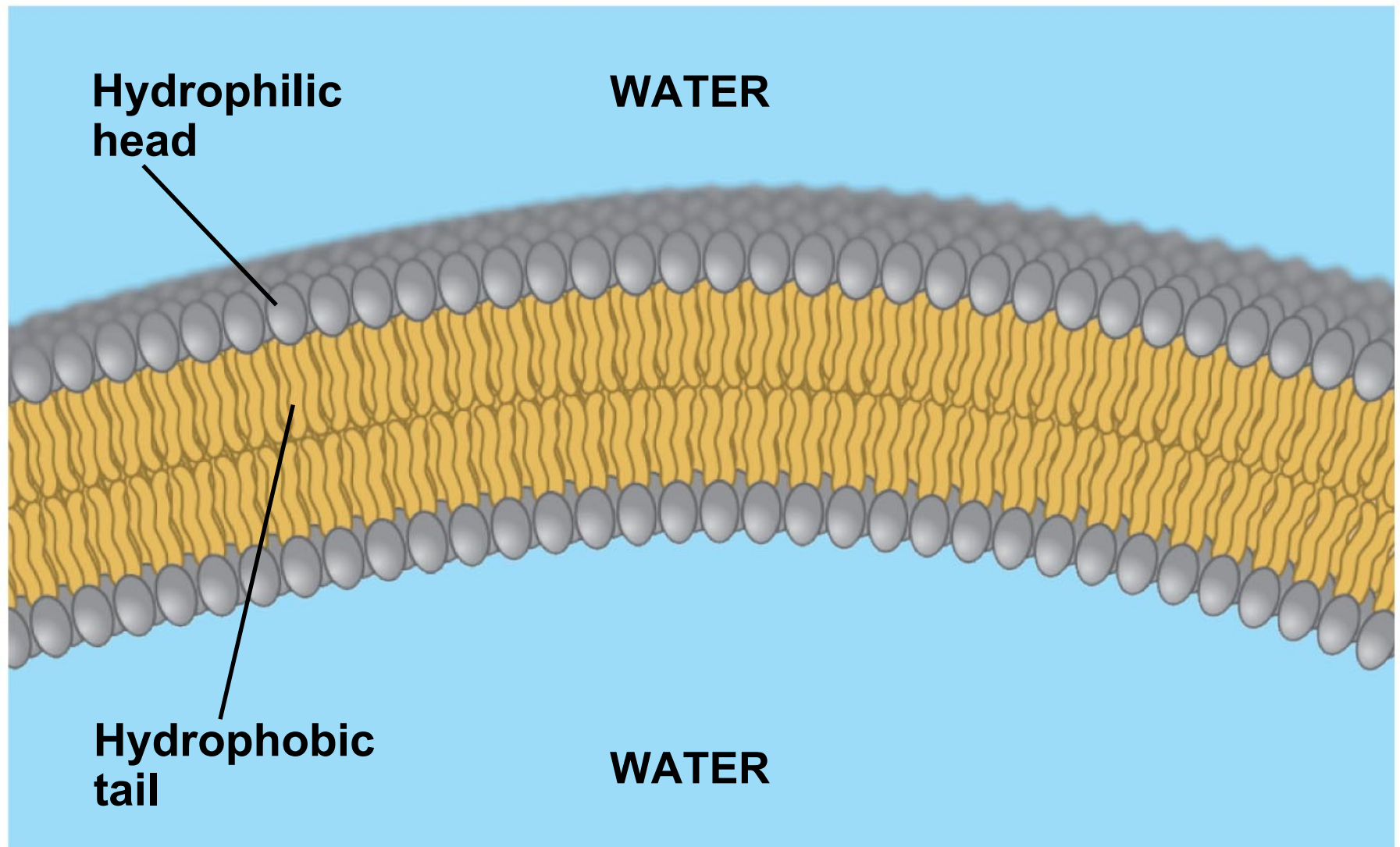


(a) Structural formula

(b) Space-filling model

-
- When phospholipids are added to water, they self-assemble into a bilayer, with the hydrophobic tails pointing toward the interior
 - The structure of phospholipids results in a bilayer arrangement found in cell membranes
 - Phospholipids are the major component of all cell membranes

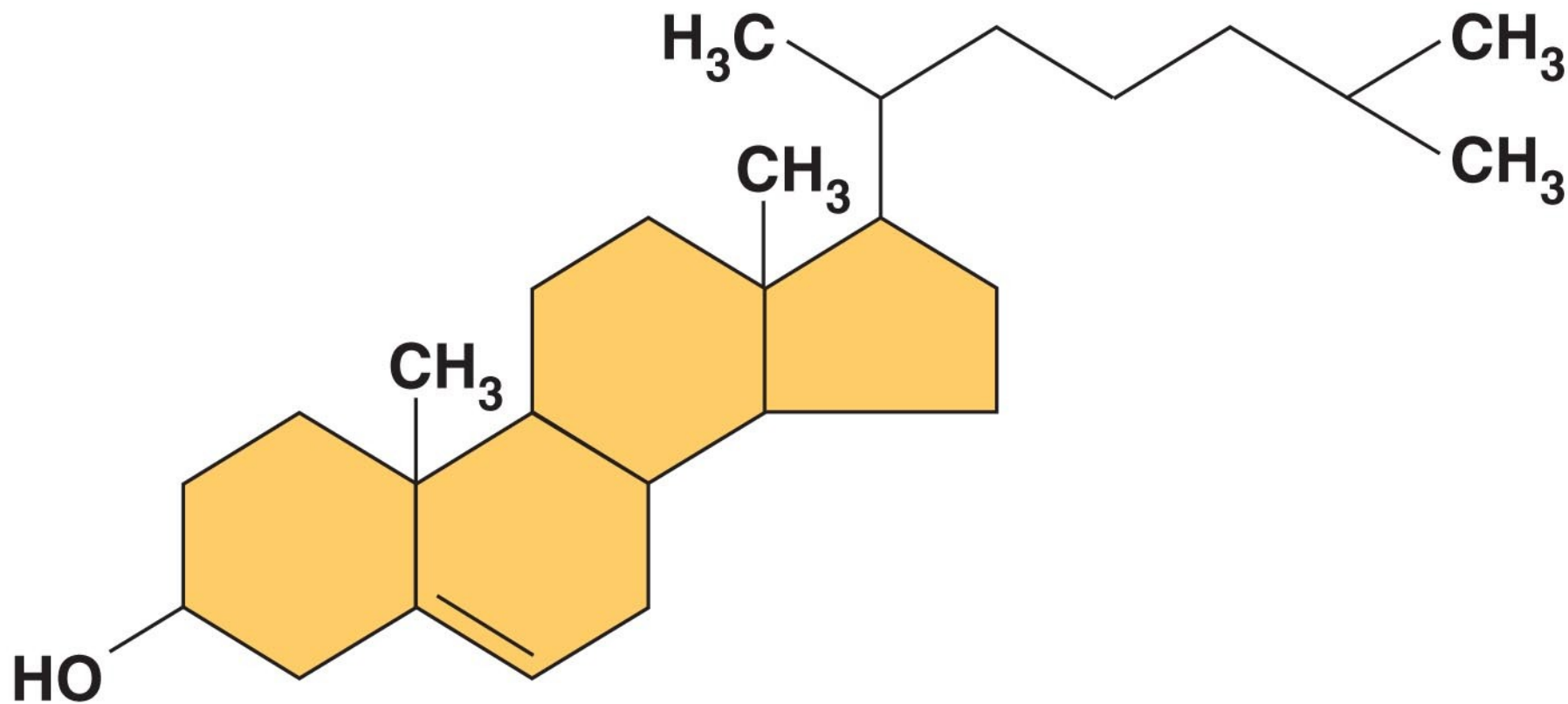
Fig. 5-14



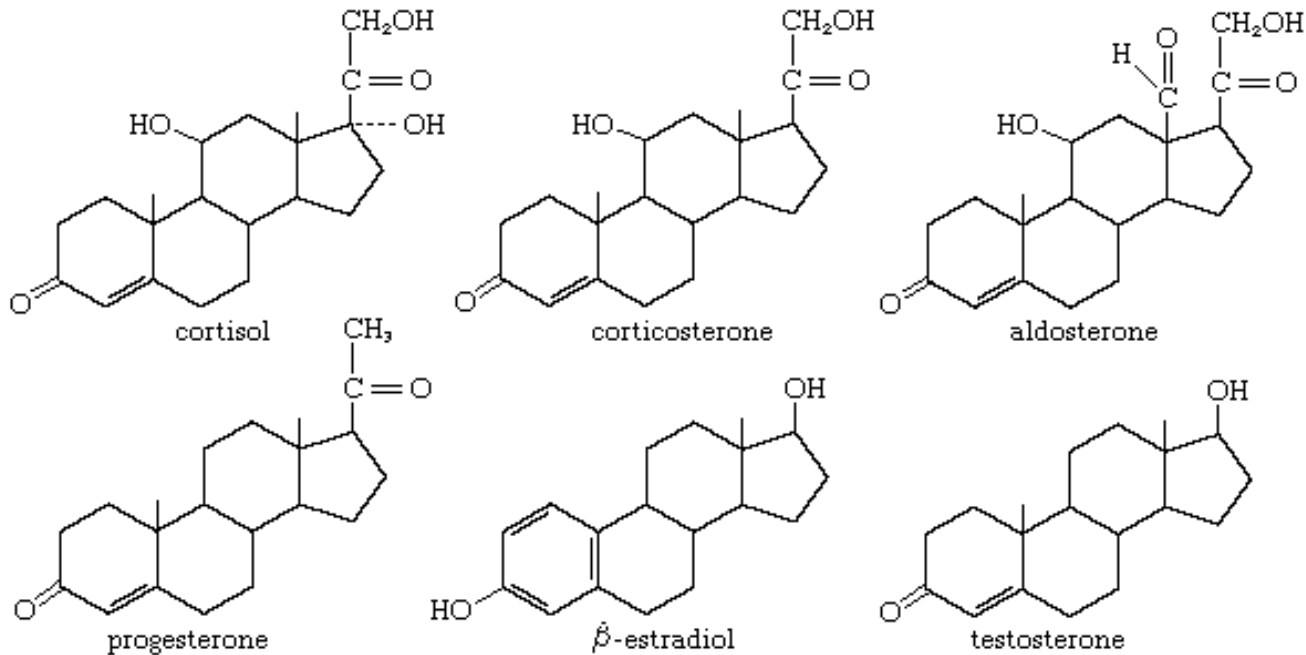
Steroids

- **Steroids** are lipids characterized by a carbon skeleton consisting of four fused rings
- **Cholesterol**, an important steroid, is a component in animal cell membranes
- Although cholesterol is essential in animals, high levels in the blood may contribute to cardiovascular disease

Fig. 5-15



Examples of Steroid Hormones



- Some of these hormones have intracellular receptors. Why?

Concept 5.4: Proteins have many structures, resulting in a wide range of functions

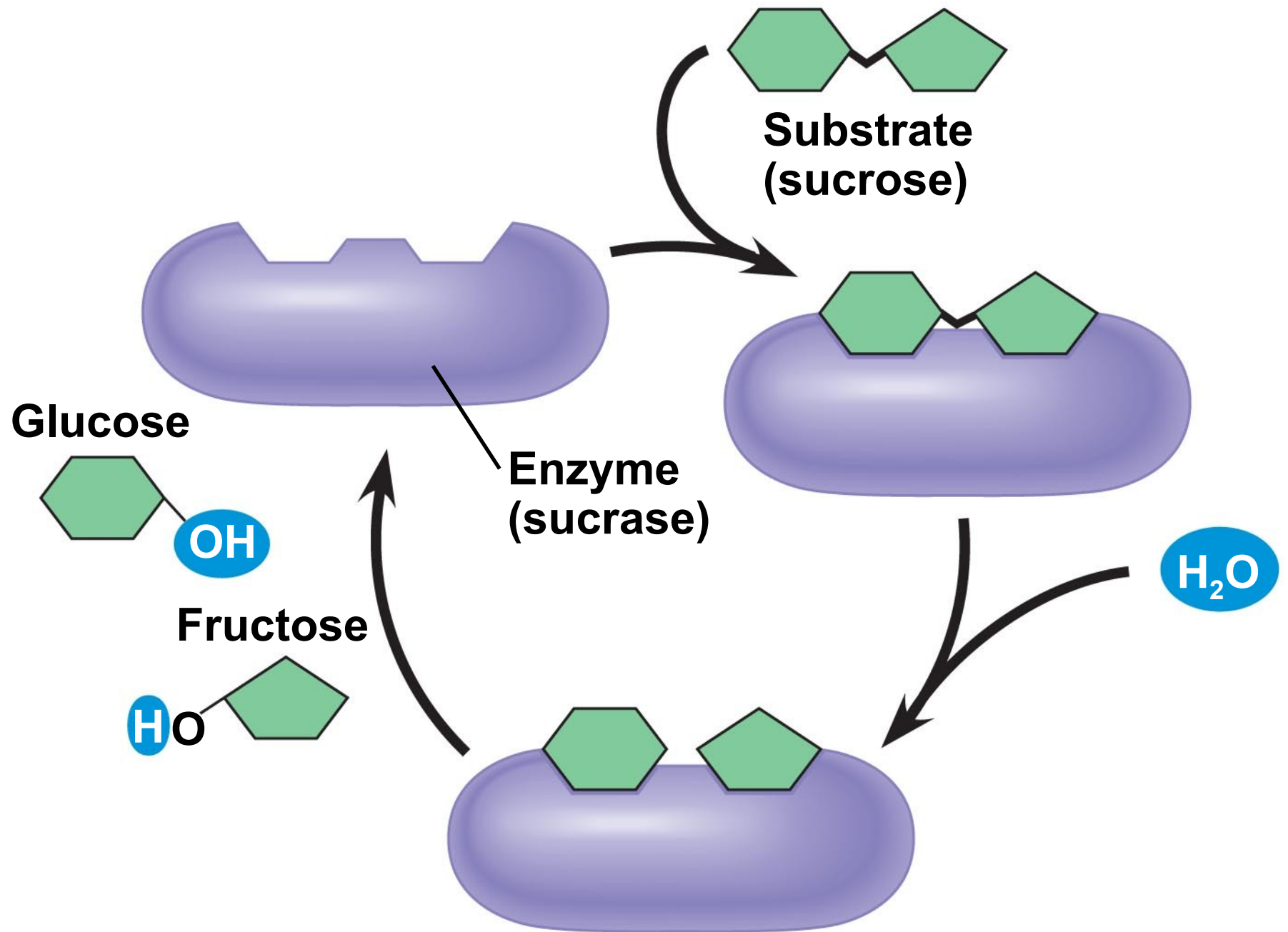
- Proteins account for more than 50% of the dry mass of most cells
- Protein functions include structural support, storage, transport, cellular communications, movement, and defense against foreign substances

Table 5.1 An Overview of Protein Functions

Type of Protein	Function	Examples
Enzymatic proteins	Selective acceleration of chemical reactions	Digestive enzymes
Structural proteins	Support	Silk fibers; collagen and elastin in animal connective tissues; keratin in hair, horns, feathers, and other skin appendages
Storage proteins	Storage of amino acids	Ovalbumin in egg white; casein, the protein of milk; storage proteins in plant seeds
Transport proteins	Transport of other substances	Hemoglobin, transport proteins
Hormonal proteins	Coordination of an organism's activities	Insulin, a hormone secreted by the pancreas
Receptor proteins	Response of cell to chemical stimuli	Receptors in nerve cell membranes
Contractile and motor proteins	Movement	Actin and myosin in muscles, proteins in cilia and flagella
Defensive proteins	Protection against disease	Antibodies combat bacteria and viruses.

-
- **Enzymes** are a type of protein that acts as a **catalyst** to speed up chemical reactions
 - Enzymes can perform their functions repeatedly, functioning as workhorses that carry out the processes of life

Fig. 5-16



Polypeptides

- **Polypeptides** are polymers built from the same set of 20 amino acids
- A **protein** consists of one or more polypeptides

Amino Acid Monomers

- **Amino acids** are organic molecules with carboxyl and amino groups
- Amino acids differ in their properties due to differing side chains, called R groups

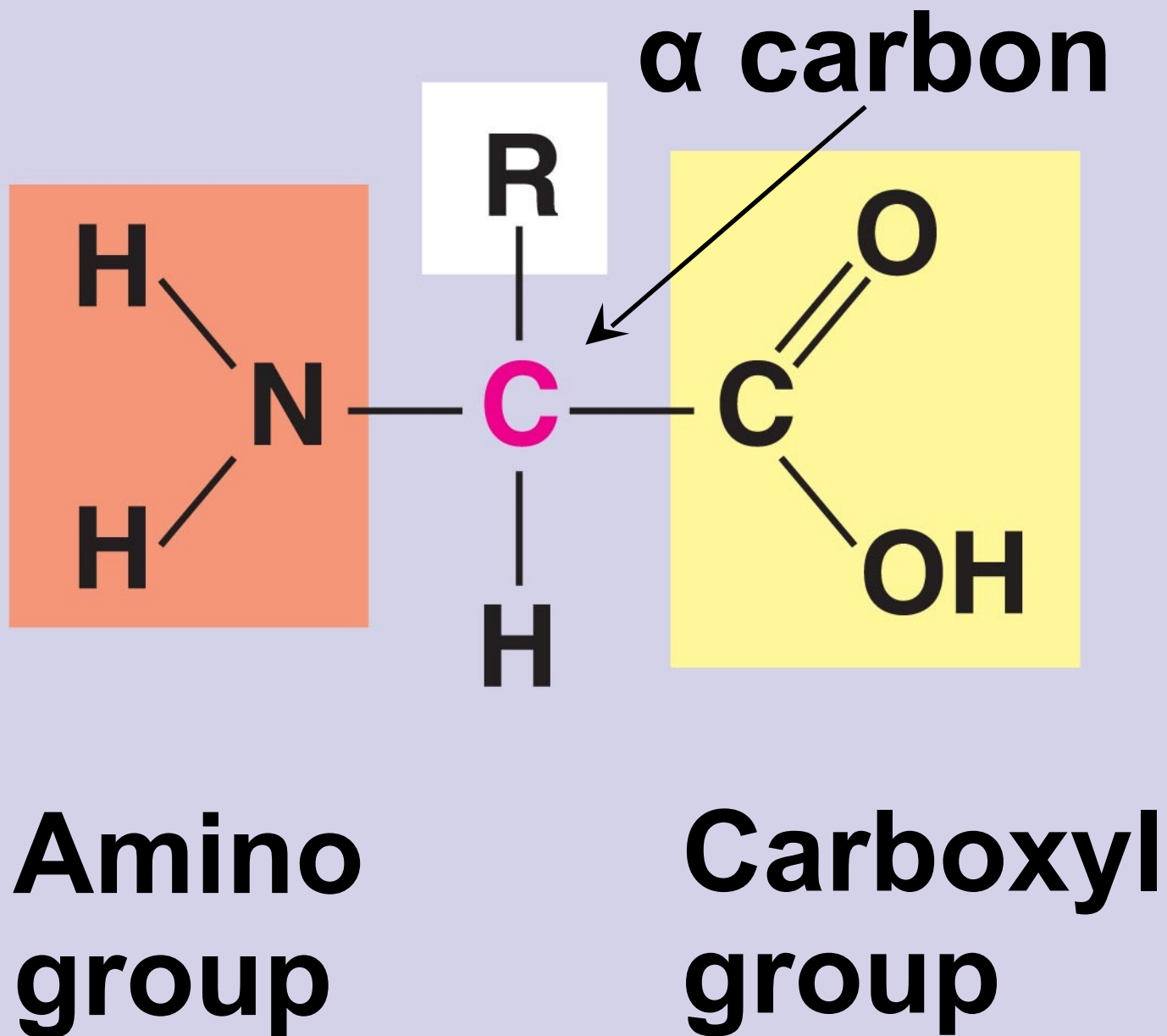
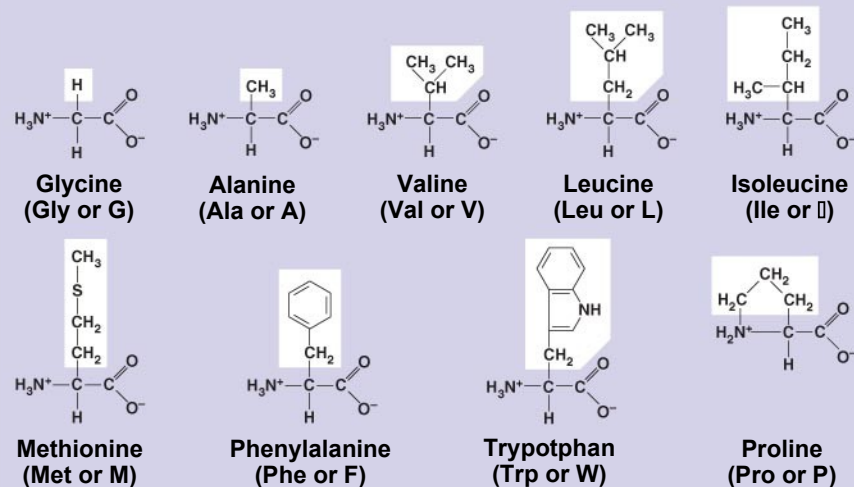
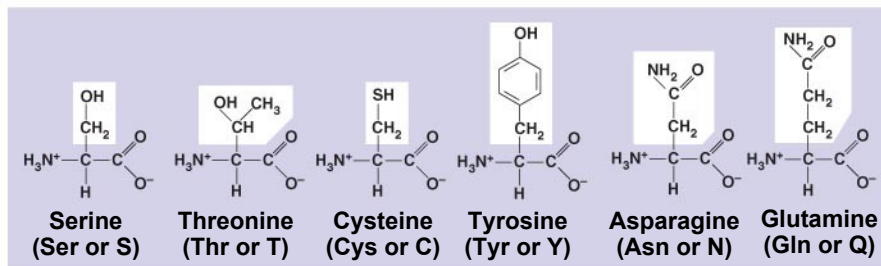


Fig. 5-17

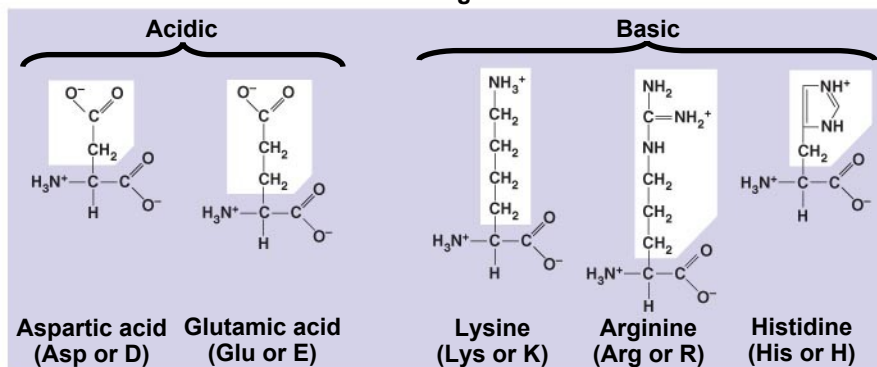
Nonpolar



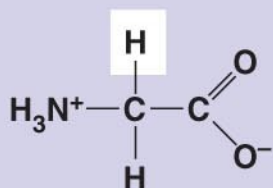
Polar



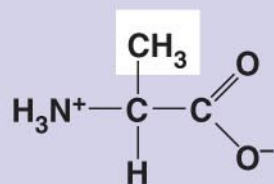
Electrically charged



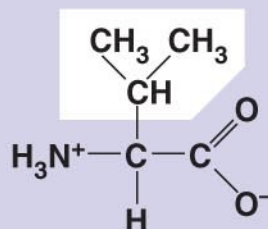
Nonpolar



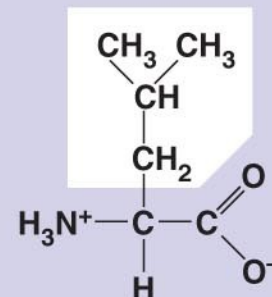
Glycine
(Gly or G)



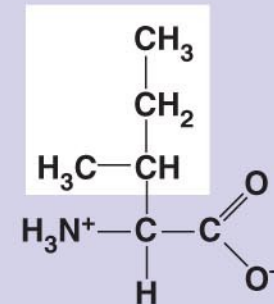
Alanine
(Ala or A)



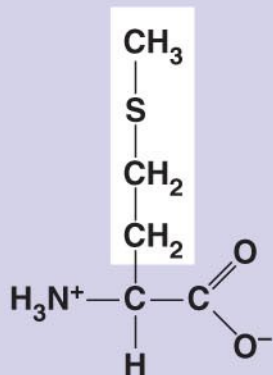
Valine
(Val or V)



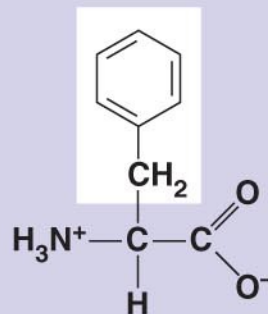
Leucine
(Leu or L)



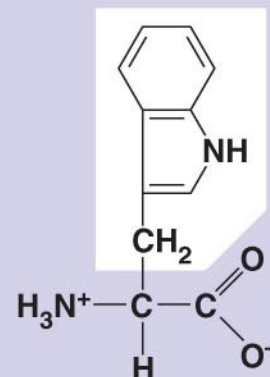
Isoleucine
(Ile or I)



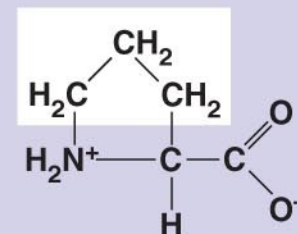
Methionine
(Met or M)



Phenylalanine
(Phe or F)

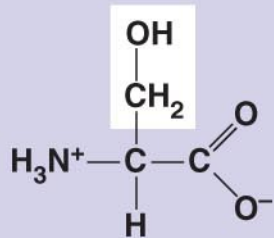


Tryptophan
(Trp or W)

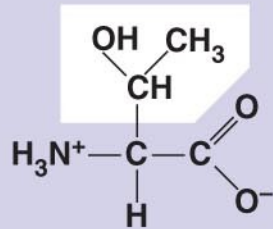


Proline
(Pro or P)

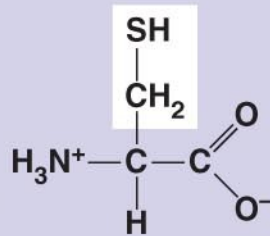
Polar



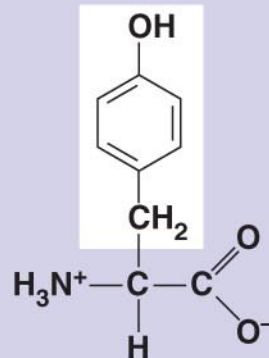
Serine
(Ser or S)



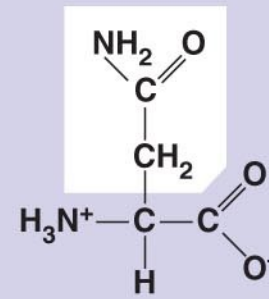
Threonine
(Thr or T)



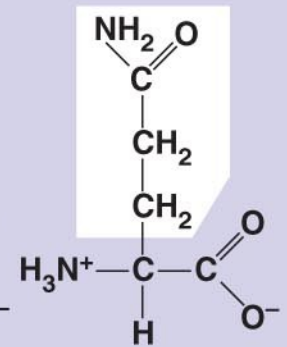
Cysteine
(Cys or C)



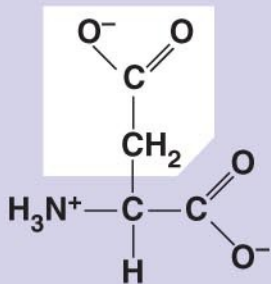
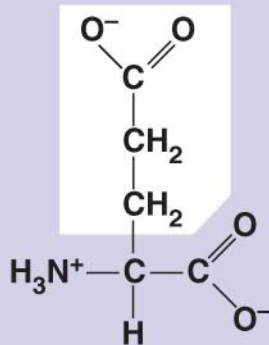
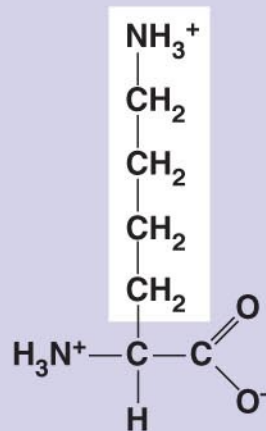
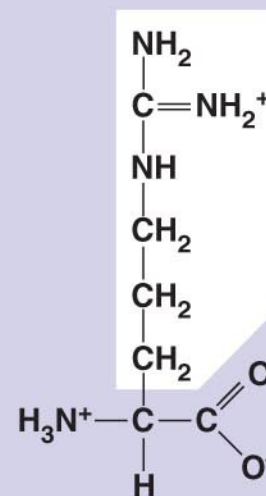
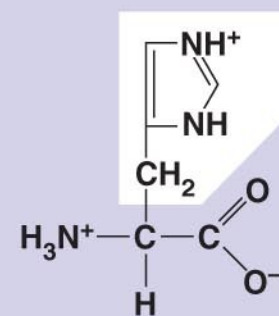
Tyrosine
(Tyr or Y)



Asparagine
(Asn or N)



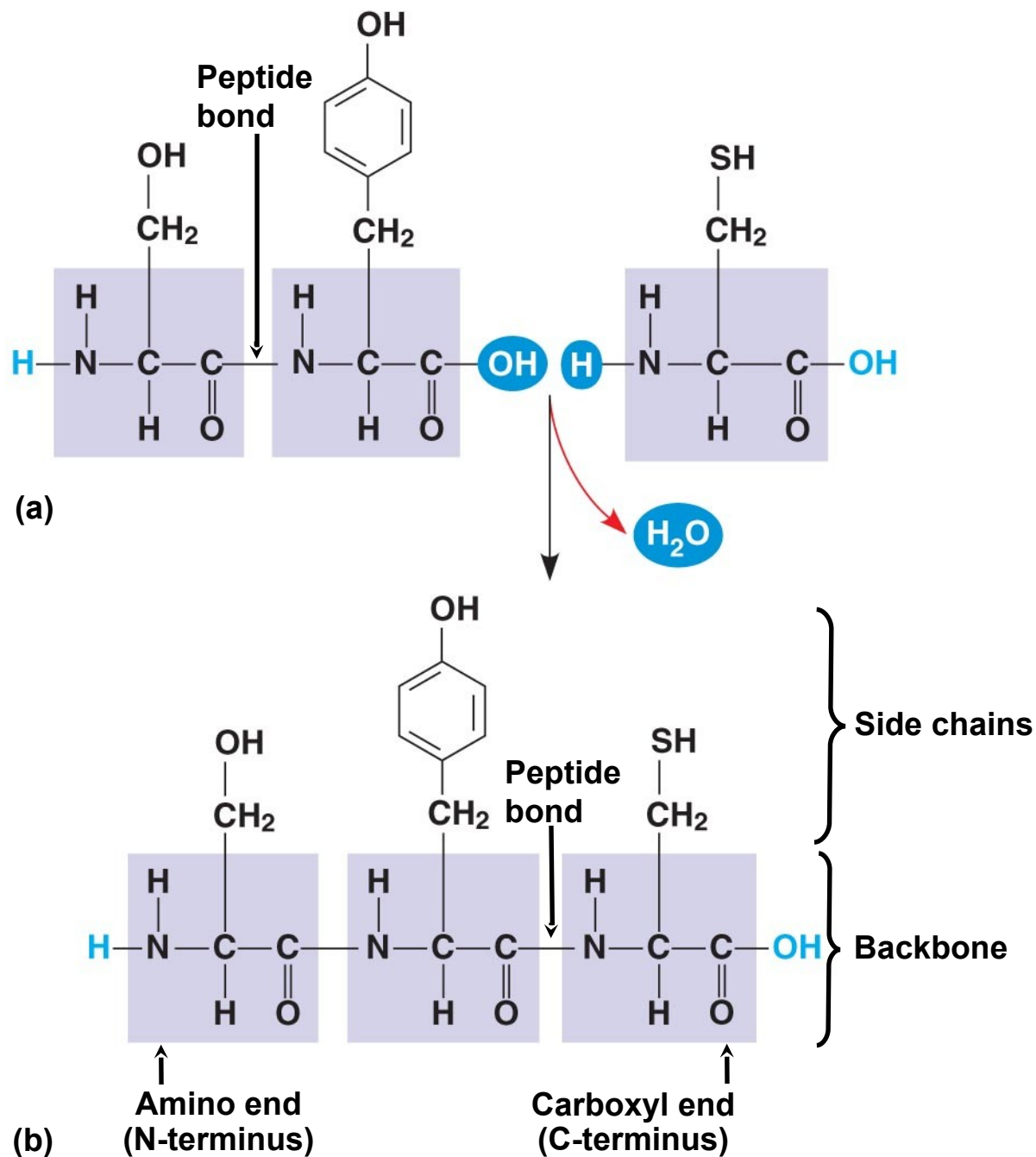
Glutamine
(Gln or Q)

Electrically charged**Acidic****Aspartic acid
(Asp or D)****Glutamic acid
(Glu or E)****Basic****Lysine
(Lys or K)****Arginine
(Arg or R)****Histidine
(His or H)**

Amino Acid Polymers

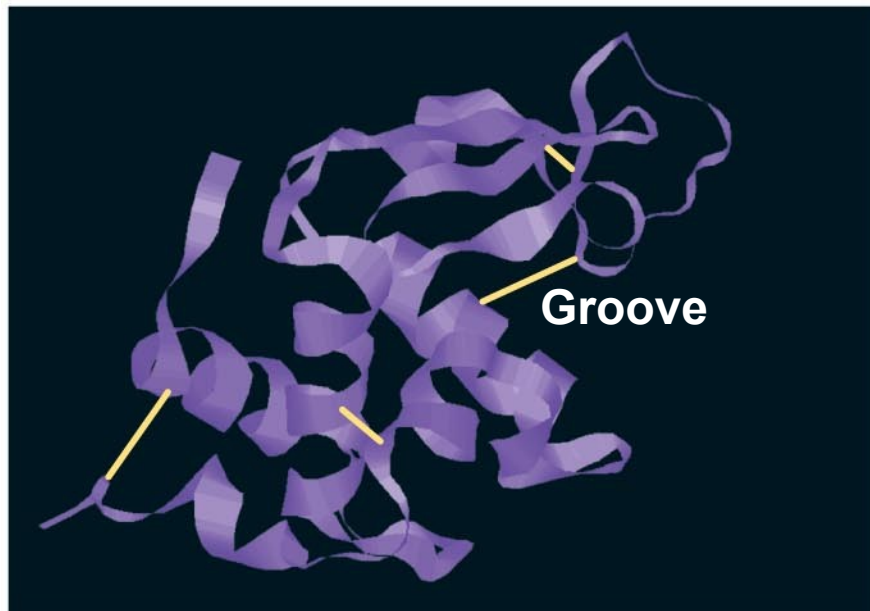
- Amino acids are linked by **peptide bonds**
- A polypeptide is a polymer of amino acids
- Polypeptides range in length from a few to more than a thousand monomers
- Each polypeptide has a unique linear sequence of amino acids

Fig. 5-18



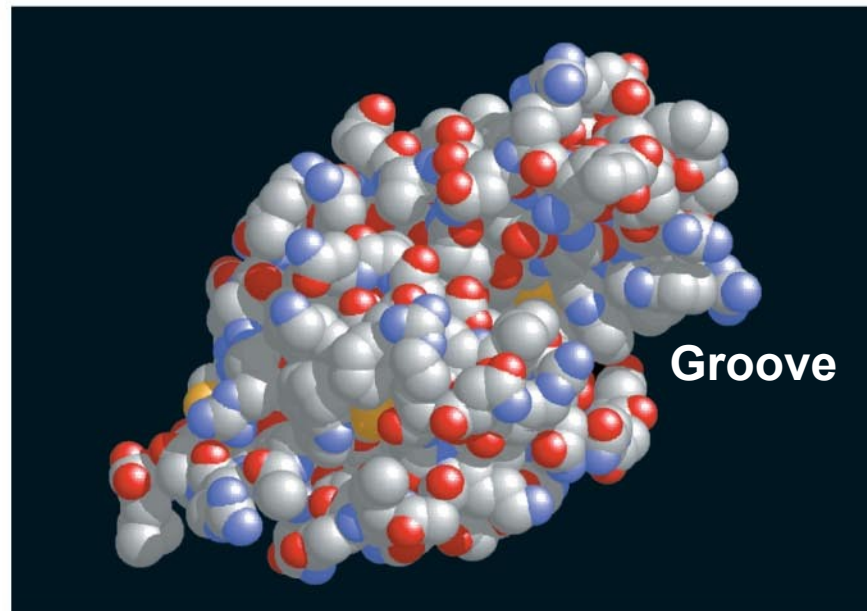
Protein Structure and Function

- A functional protein consists of one or more polypeptides twisted, folded, and coiled into a unique shape



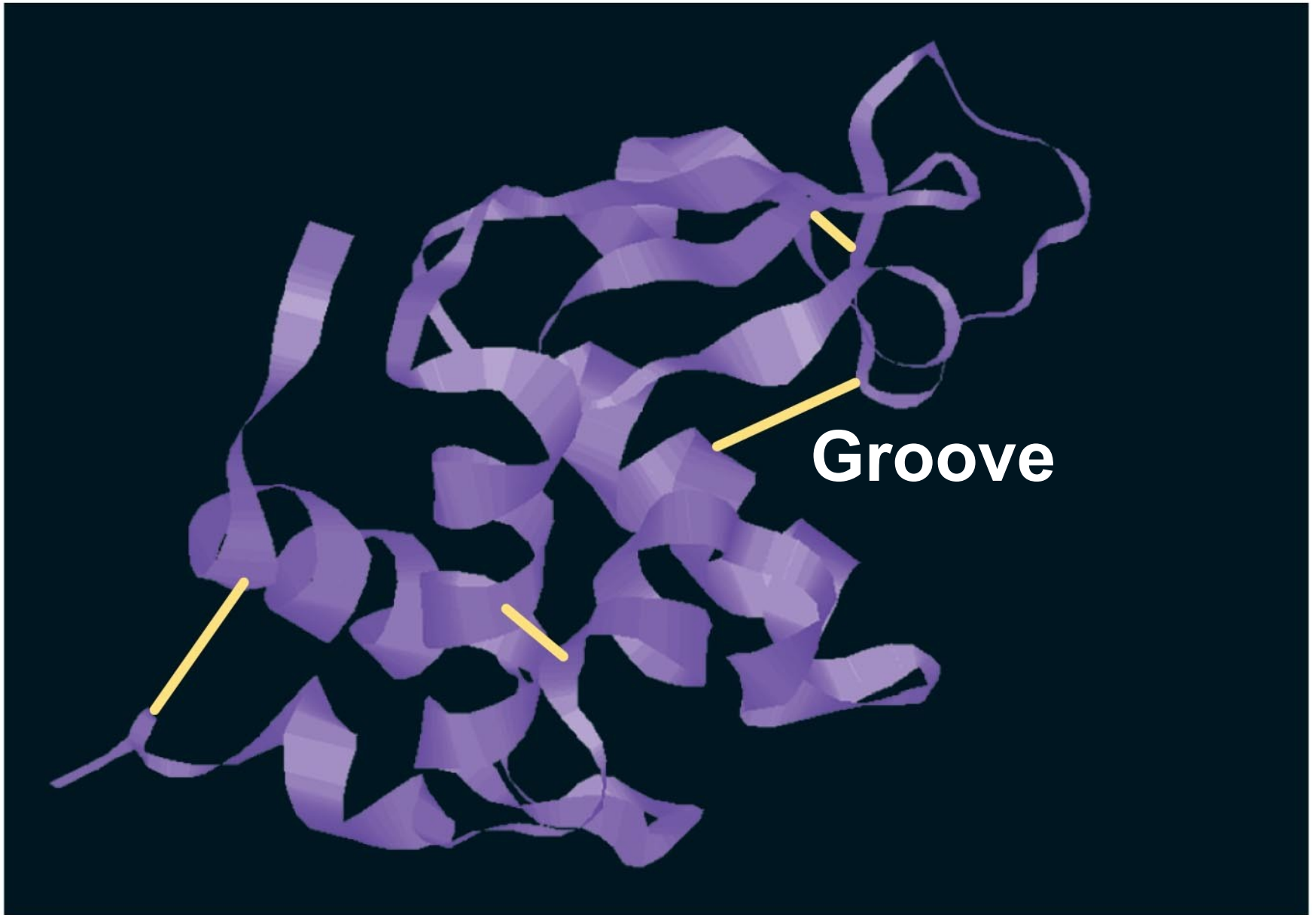
(a) A ribbon model of lysozyme

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

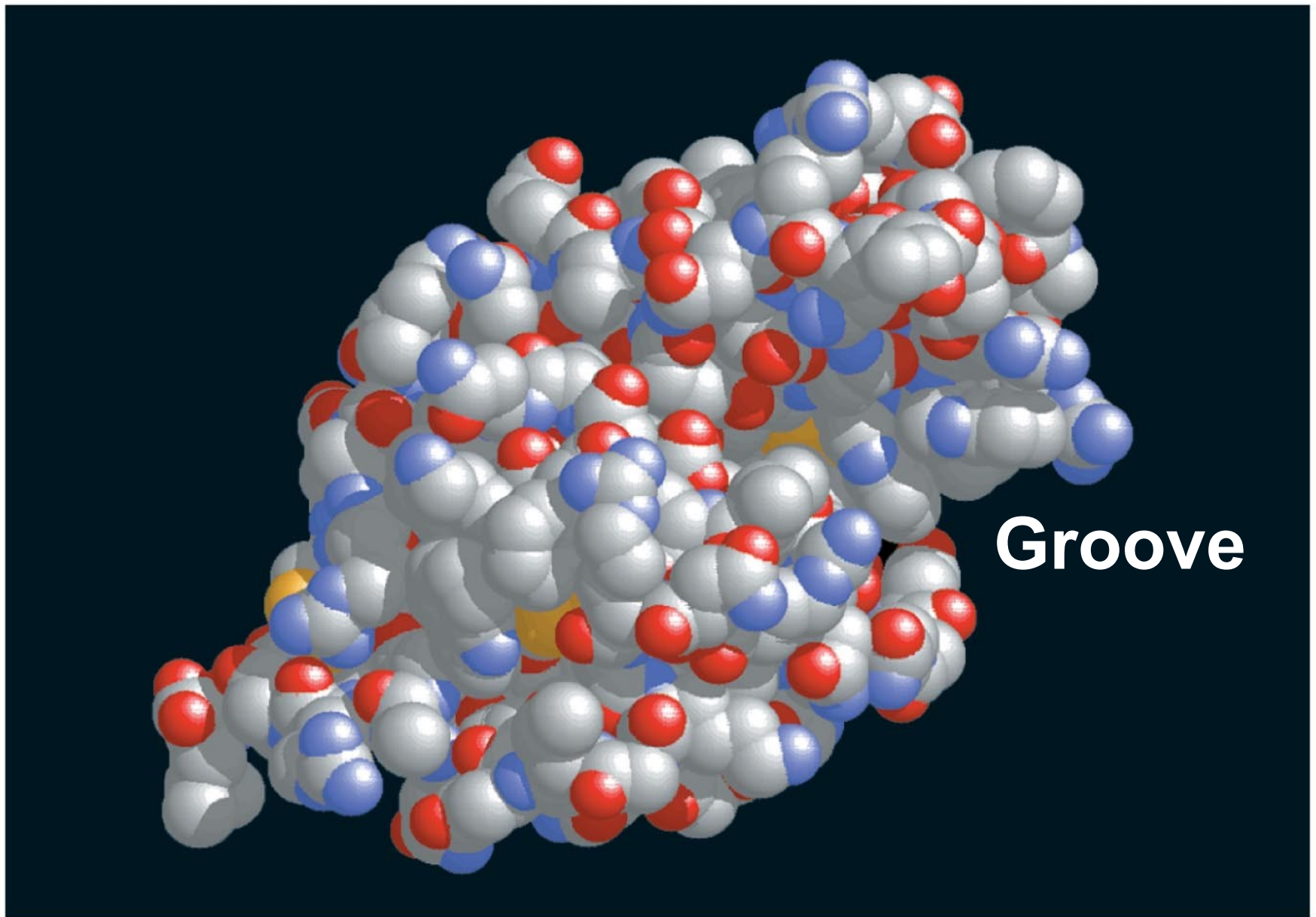


(b) A space-filling model of lysozyme

Fig. 5-19a



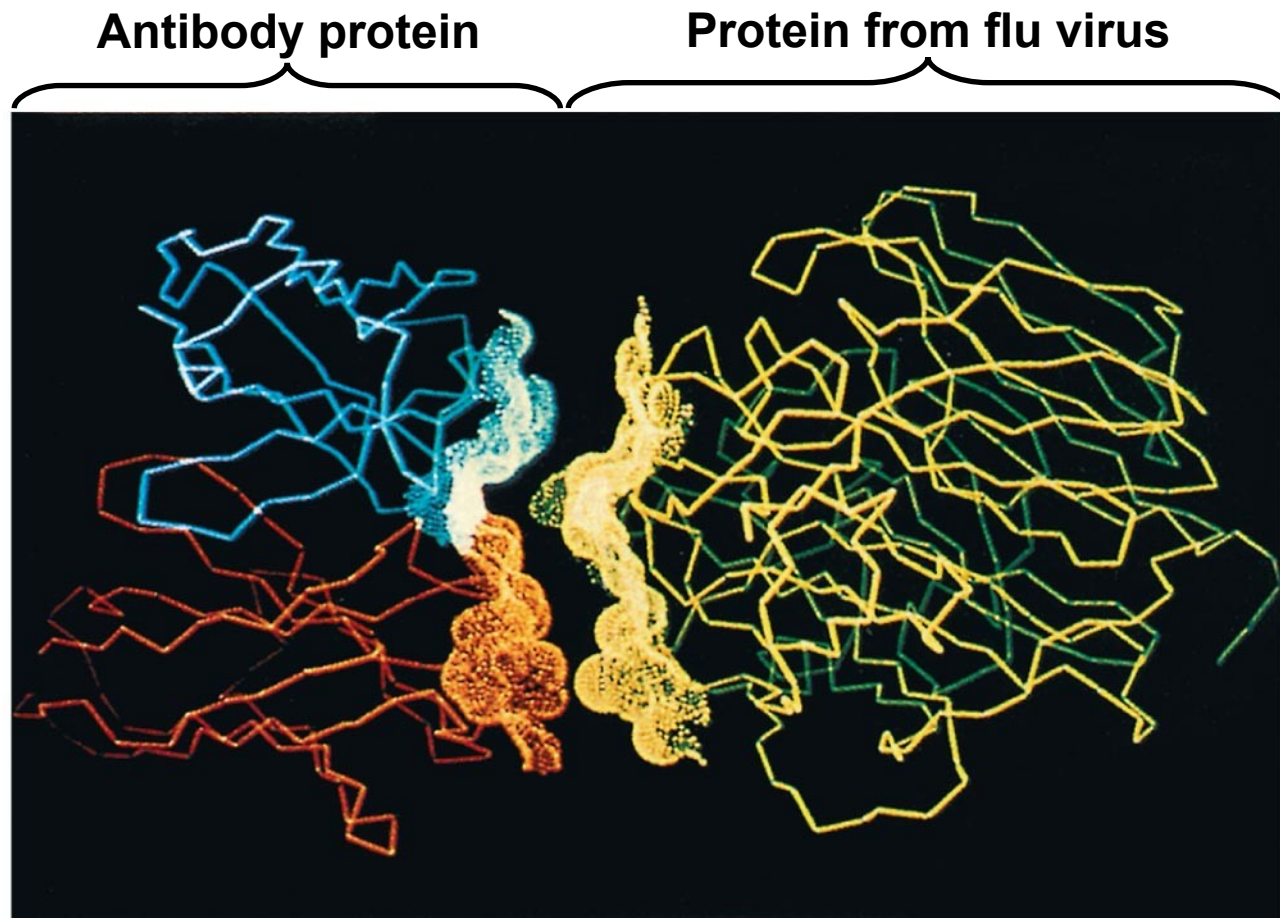
(a) A ribbon model of lysozyme



(b) A space-filling model of lysozyme

-
- The sequence of amino acids determines a protein's three-dimensional structure
 - A protein's structure determines its function

Fig. 5-20



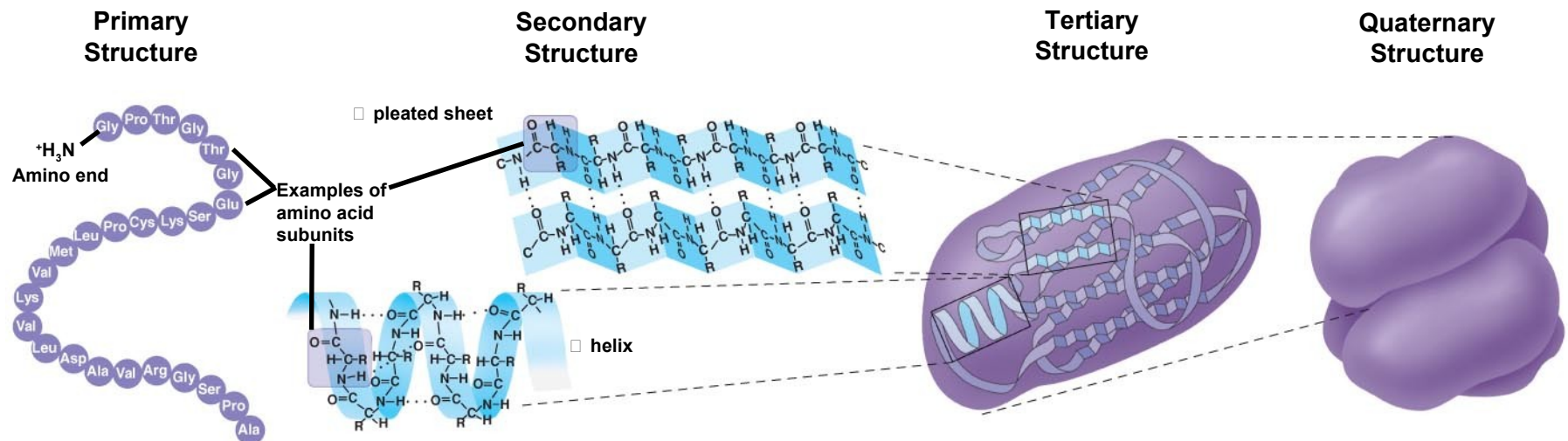
Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

Four Levels of Protein Structure

- The primary structure of a protein is its unique sequence of amino acids
- Secondary structure, found in most proteins, consists of coils and folds in the polypeptide chain
- Tertiary structure is determined by interactions among various side chains (R groups)
- Quaternary structure results when a protein consists of multiple polypeptide chains

-
- **Primary structure**, the sequence of amino acids in a protein, is like the order of letters in a long word
 - Primary structure is determined by inherited genetic information

Fig. 5-21



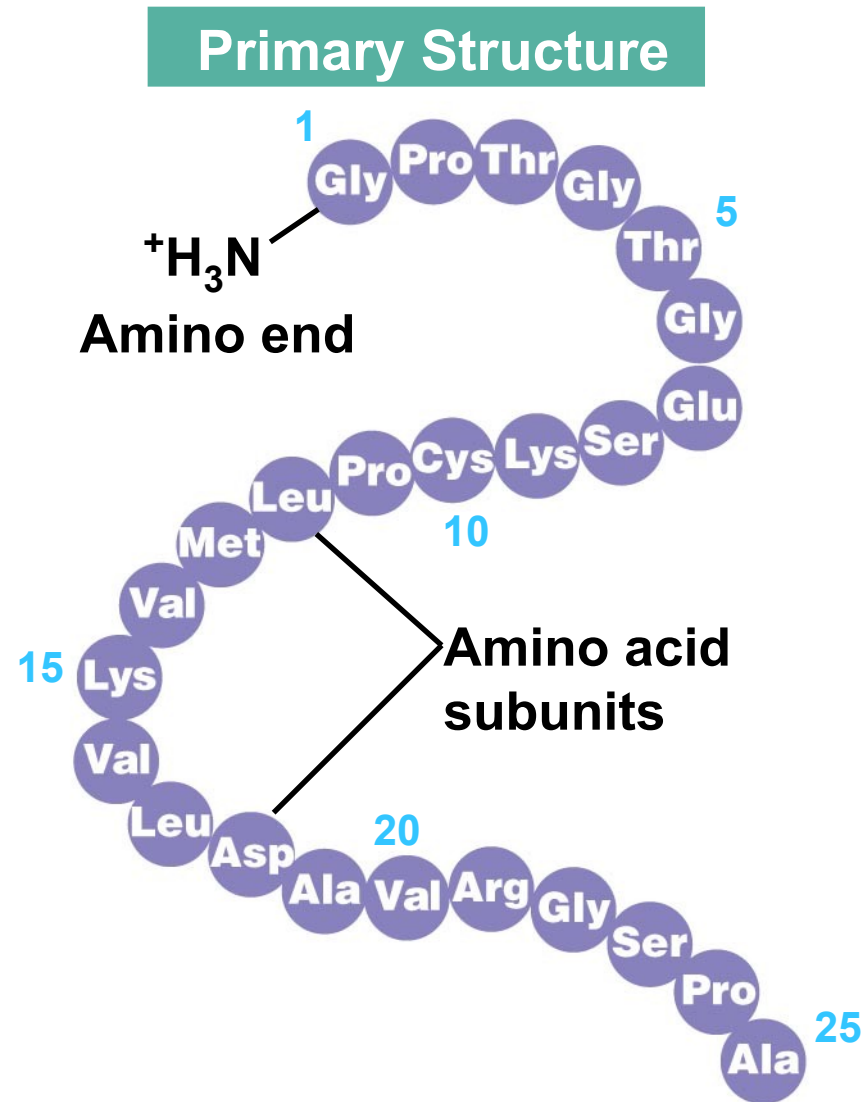
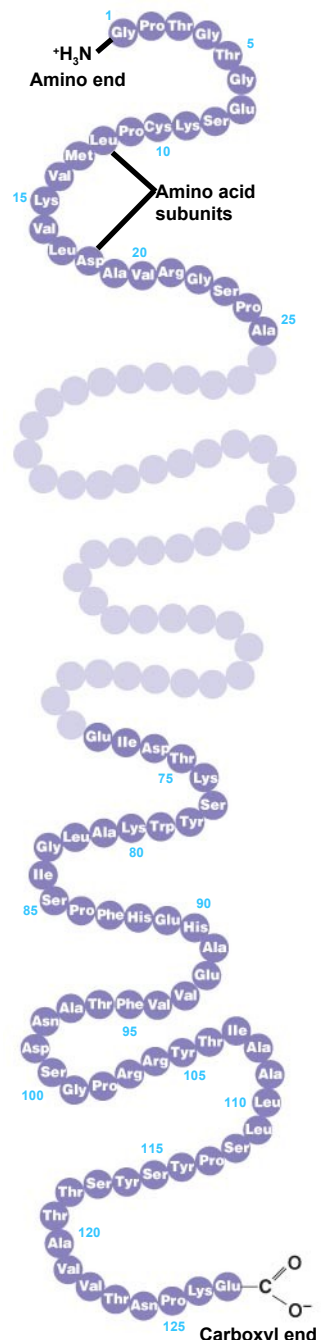


Fig. 5-21b

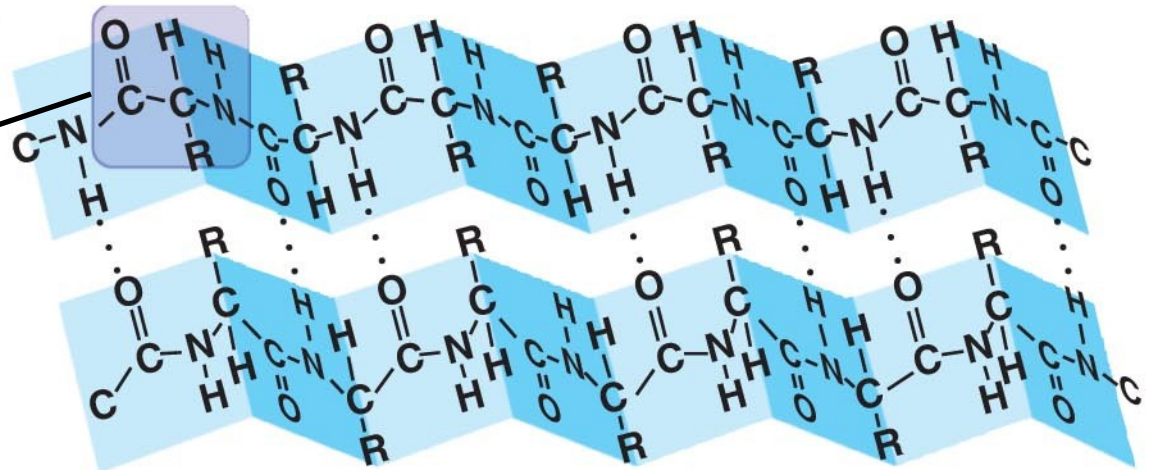


-
- The coils and folds of **secondary structure** result from hydrogen bonds between repeating constituents of the polypeptide backbone
 - Typical secondary structures are a coil called an **α helix** and a folded structure called a **β pleated sheet**

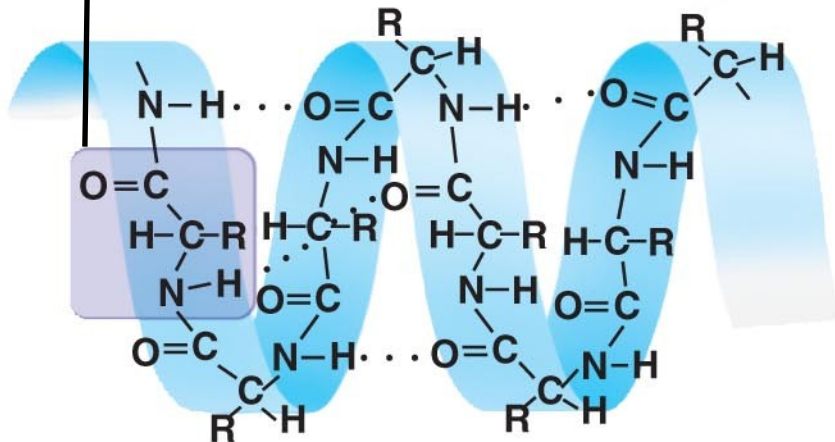
Secondary Structure

β pleated sheet

Examples of
amino acid
subunits



α helix



Abdominal glands of the spider secrete silk fibers made of a structural protein containing β pleated sheets.

The radiating strands, made of dry silk fibers, maintain the shape of the web.

The spiral strands (capture strands) are elastic, stretching in response to wind, rain, and the touch of insects.



-
- **Tertiary structure** is determined by interactions between R groups, rather than interactions between backbone constituents
 - These interactions between R groups include hydrogen bonds, ionic bonds, **hydrophobic interactions**, and van der Waals interactions
 - Strong covalent bonds called **disulfide bridges** may reinforce the protein's structure

Tertiary Structure

Quaternary Structure

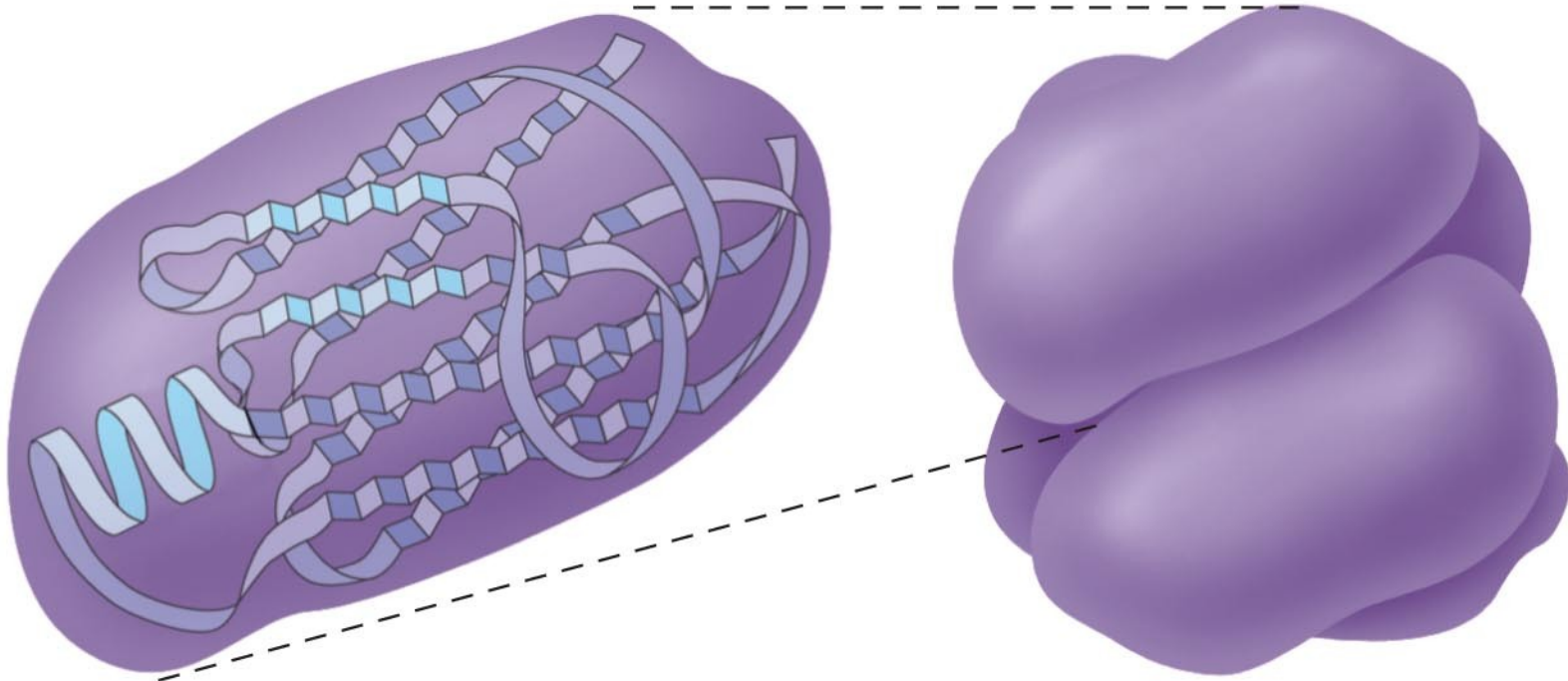
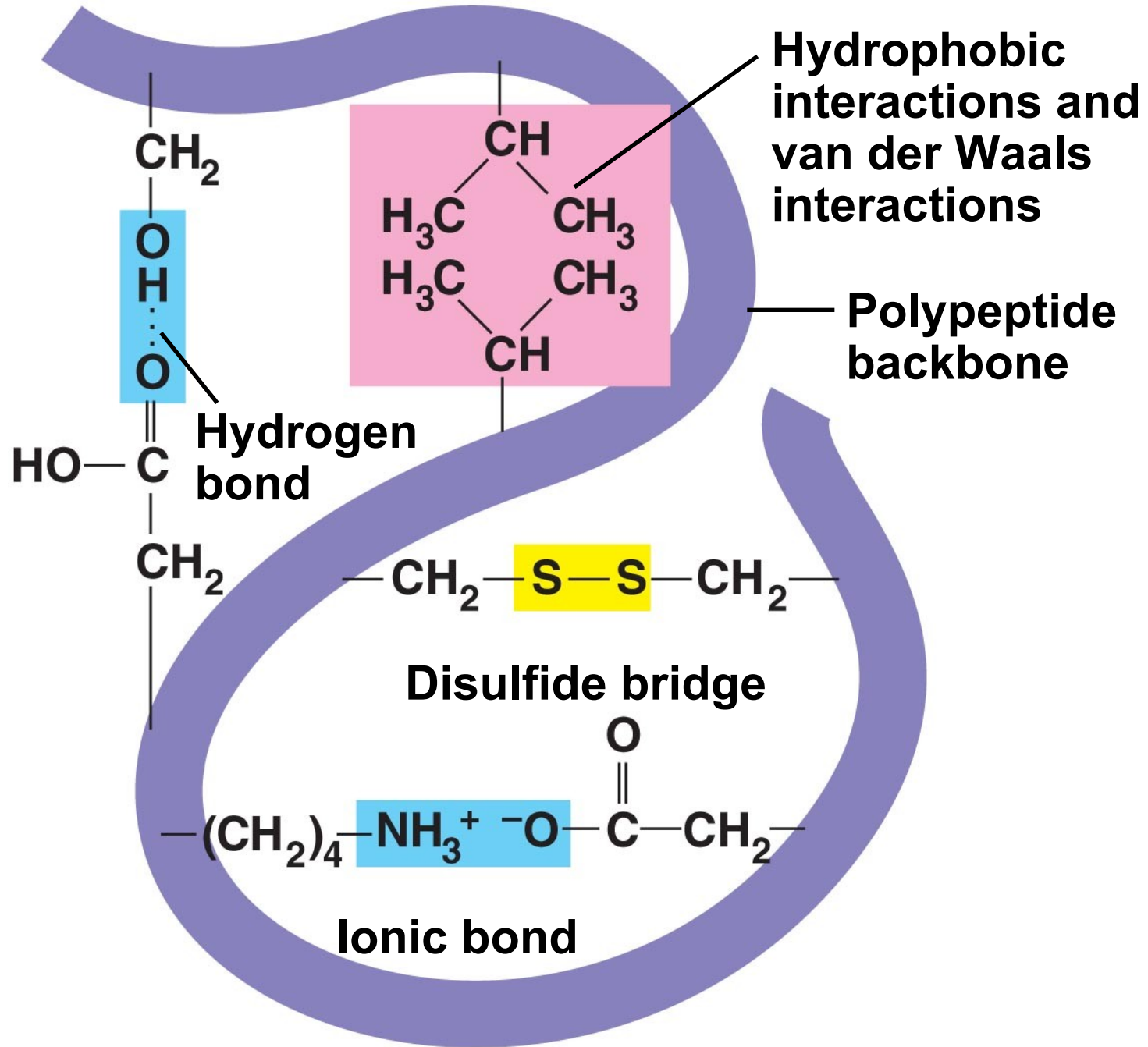
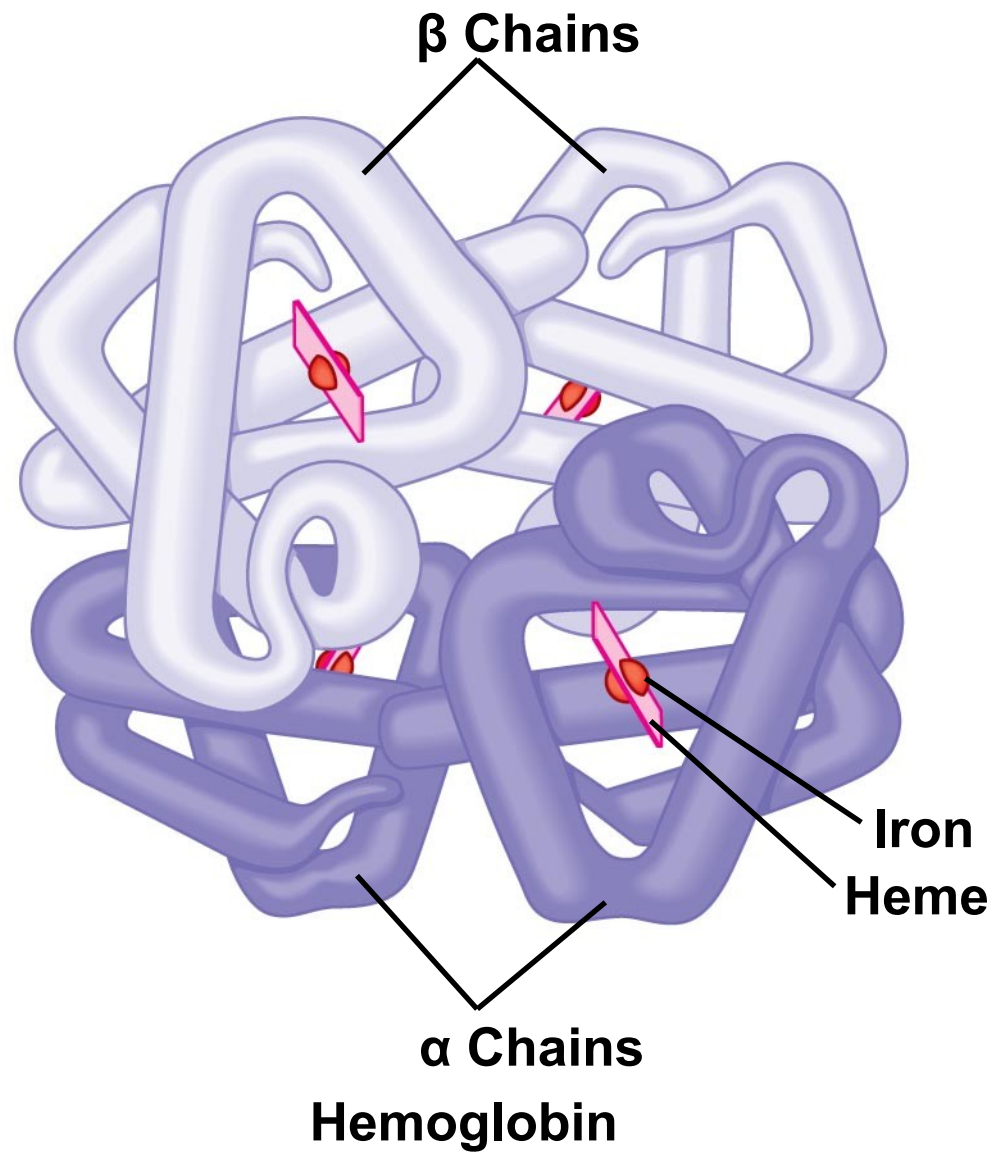


Fig. 5-21f





-
- **Quaternary structure** results when two or more polypeptide chains form one macromolecule
 - Collagen is a fibrous protein consisting of three polypeptides coiled like a rope
 - Hemoglobin is a globular protein consisting of four polypeptides: two alpha and two beta chains

Sickle-Cell Disease: A Change in Primary Structure

- A slight change in primary structure can affect a protein's structure and ability to function
- Sickle-cell disease, an inherited blood disorder, results from a single amino acid substitution in the protein hemoglobin

Fig. 5-22

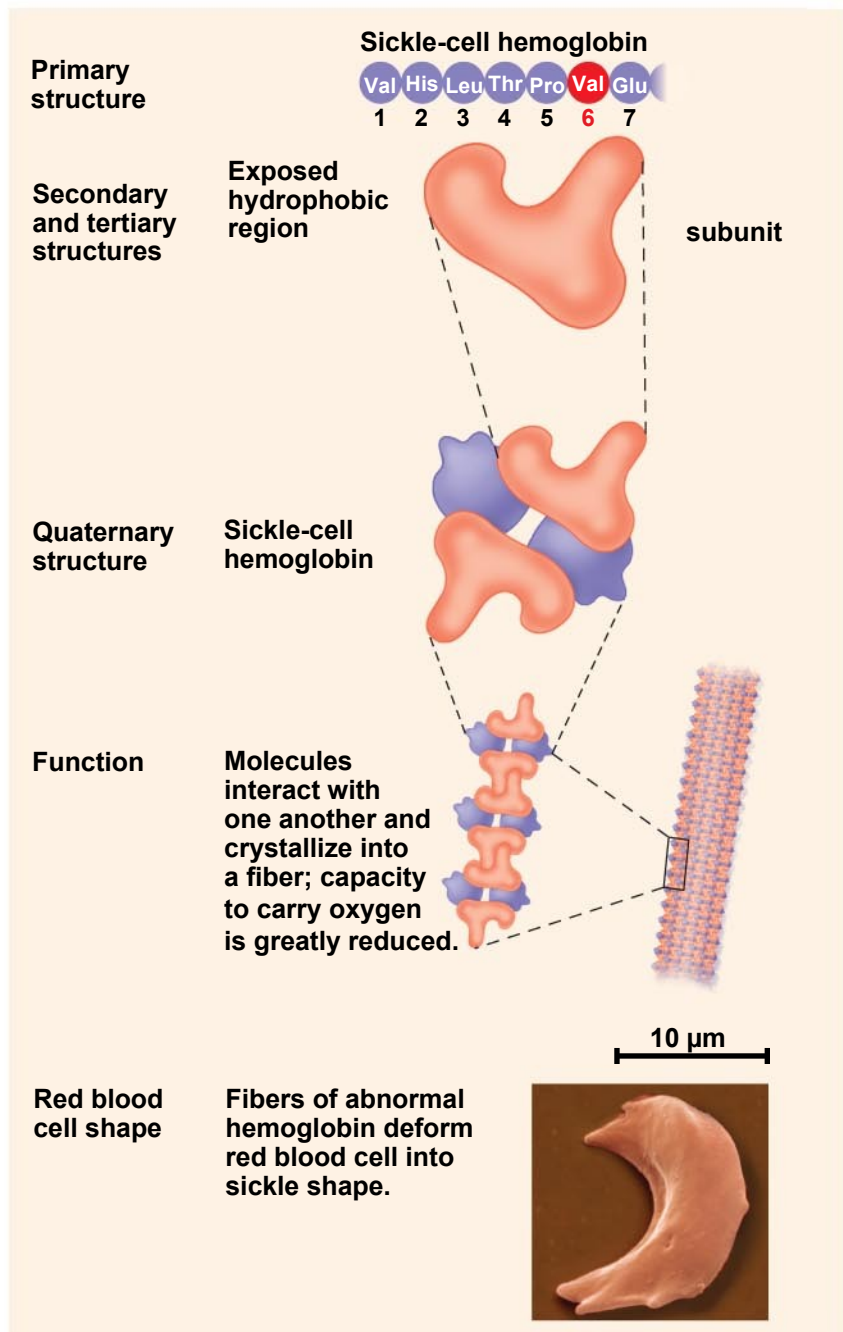
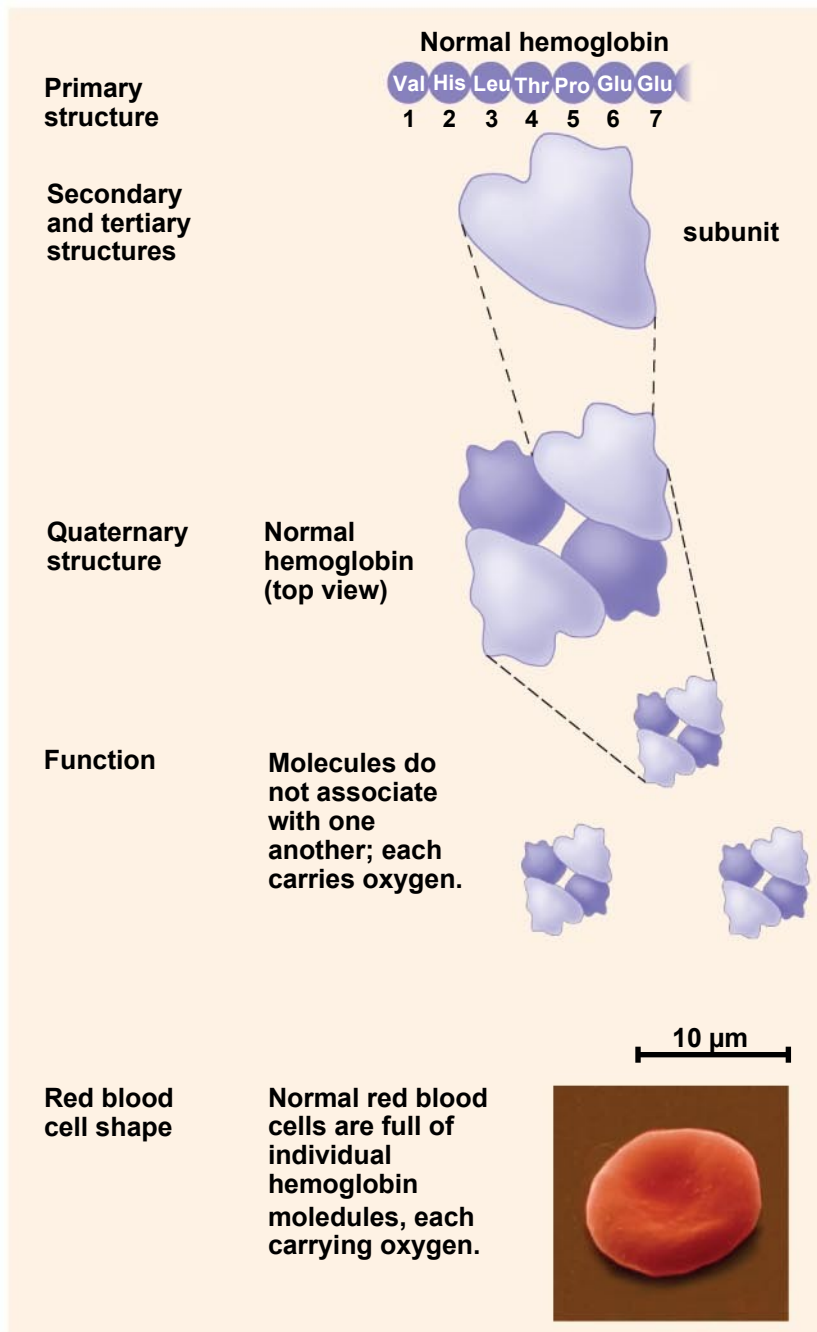
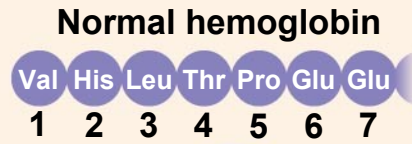
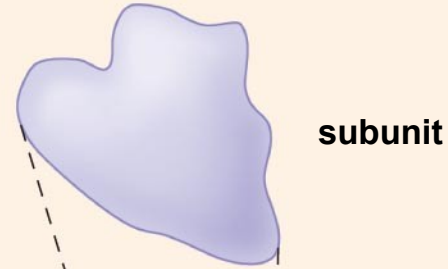


Fig. 5-22a

**Primary
structure**

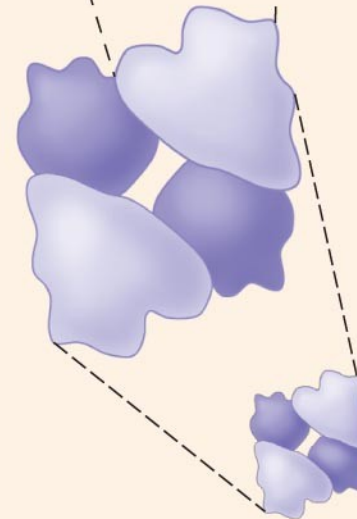


**Secondary
and tertiary
structures**



**Quaternary
structure**

**Normal
hemoglobin
(top view)**

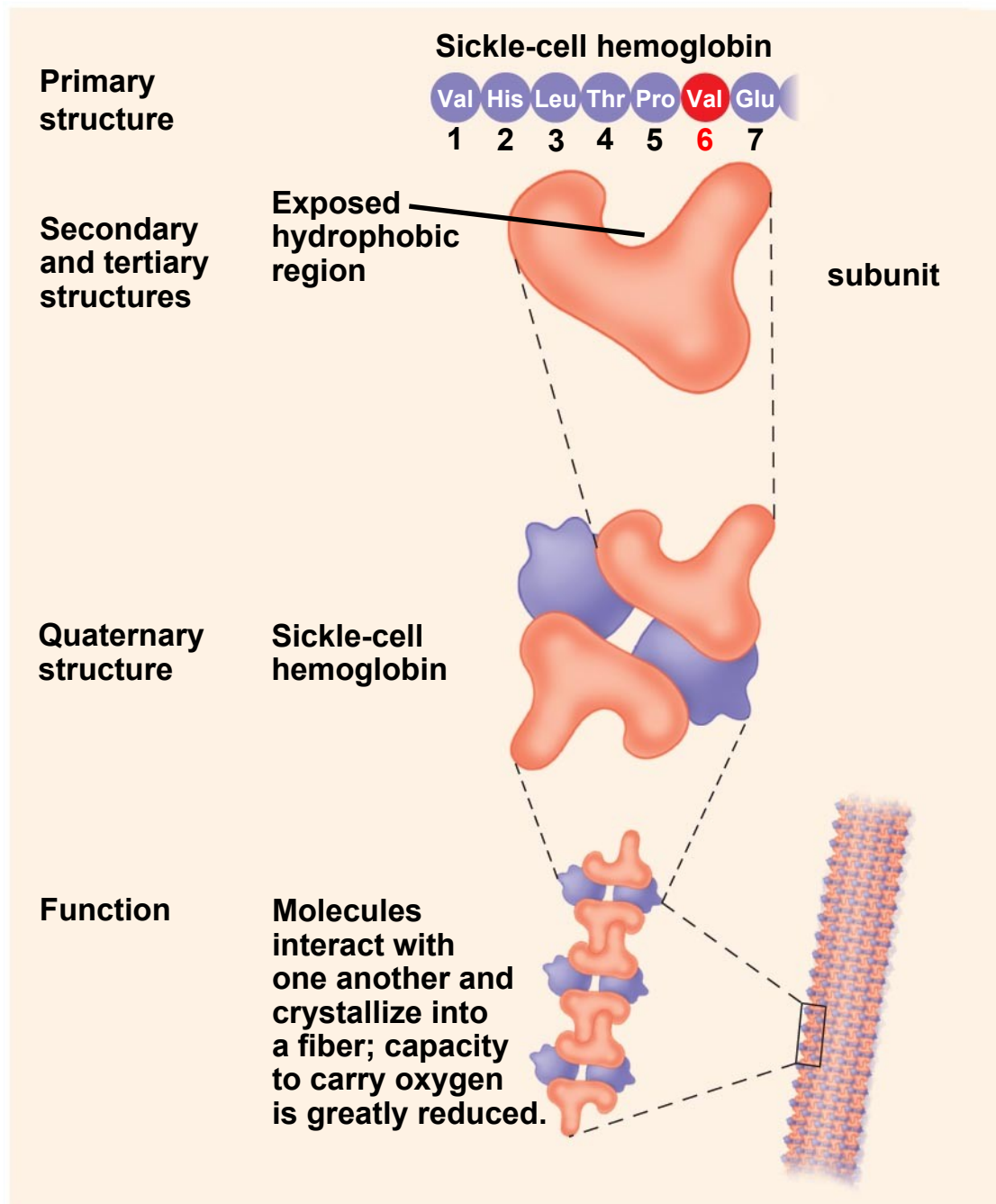


Function

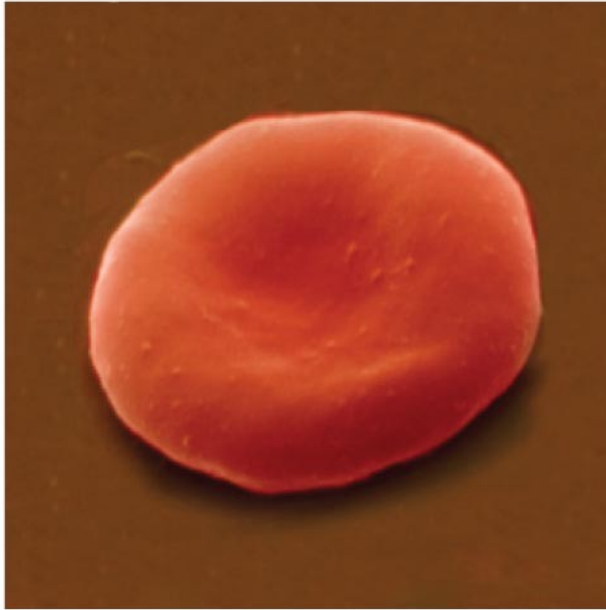
**Molecules do
not associate
with one
another; each
carries oxygen.**



Fig. 5-22b

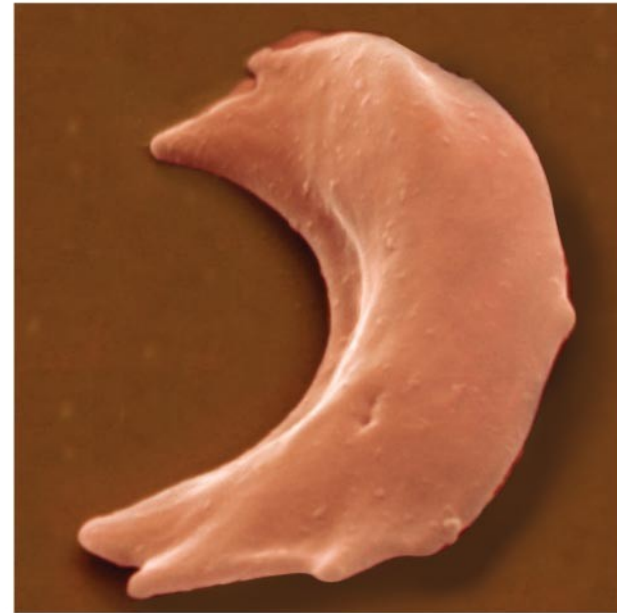


10 μm



Normal red blood cells are full of individual hemoglobin molecules, each carrying oxygen.

10 μm

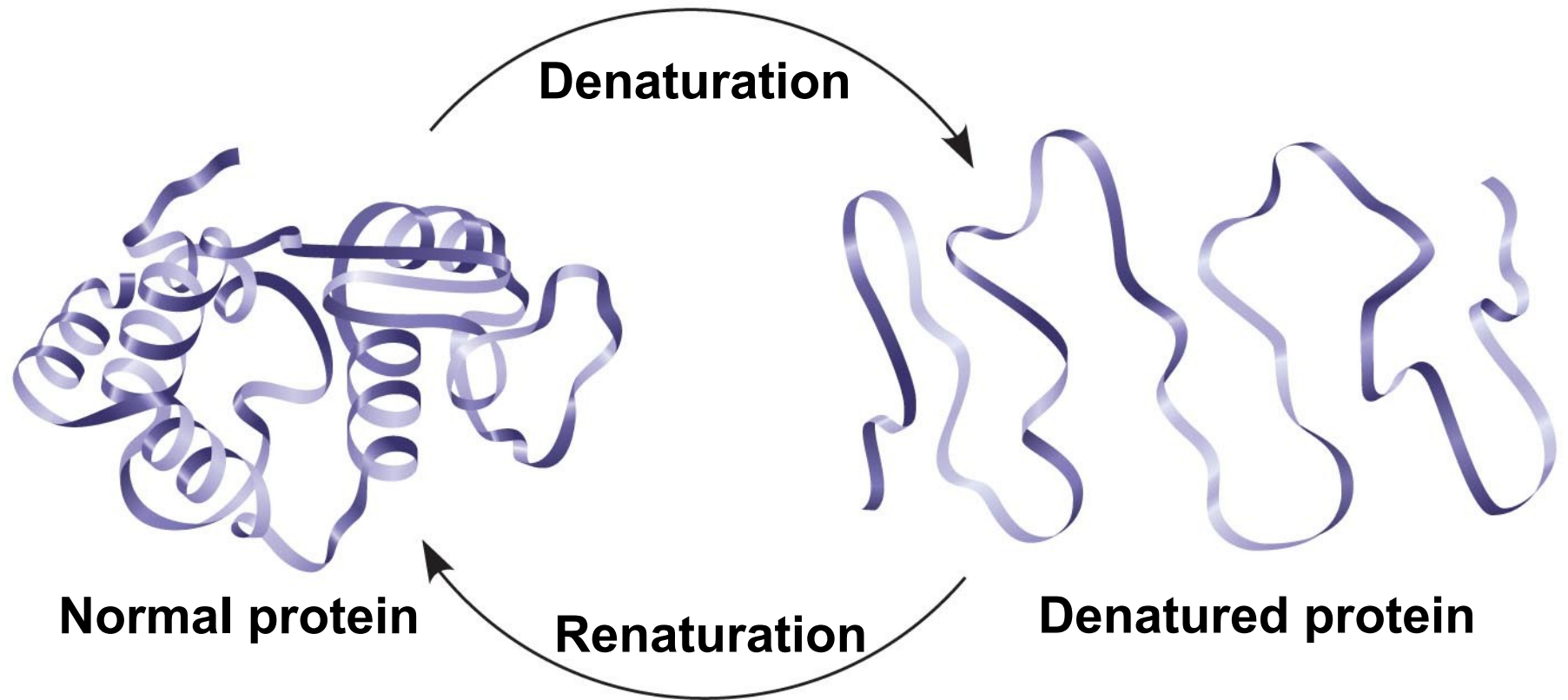


Fibers of abnormal hemoglobin deform red blood cell into sickle shape.

What Determines Protein Structure?

- In addition to primary structure, physical and chemical conditions can affect structure
- Alterations in pH, salt concentration, temperature, or other environmental factors can cause a protein to unravel
- This loss of a protein's native structure is called **denaturation**
- A denatured protein is biologically inactive

Fig. 5-23



Protein Folding in the Cell

- It is hard to predict a protein's structure from its primary structure
- Most proteins probably go through several states on their way to a stable structure
- **Chaperonins** are protein molecules that assist the proper folding of other proteins

Fig. 5-24

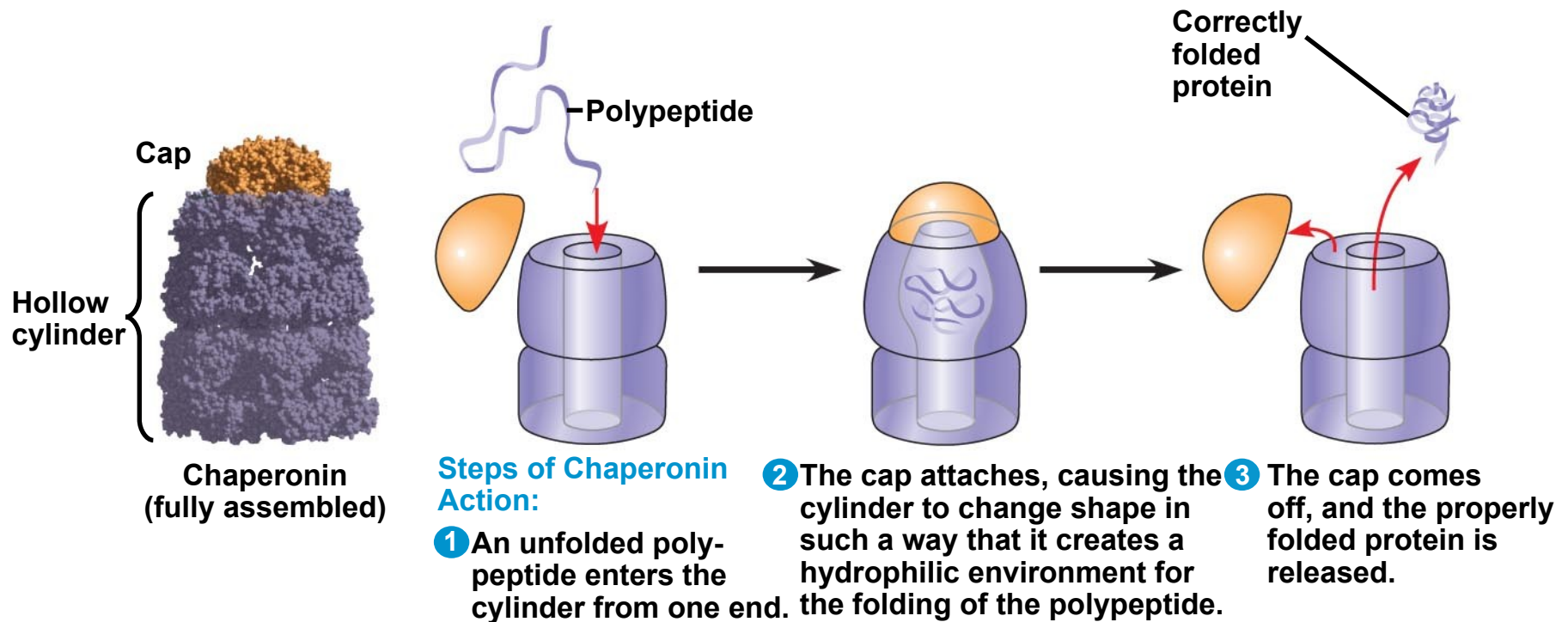
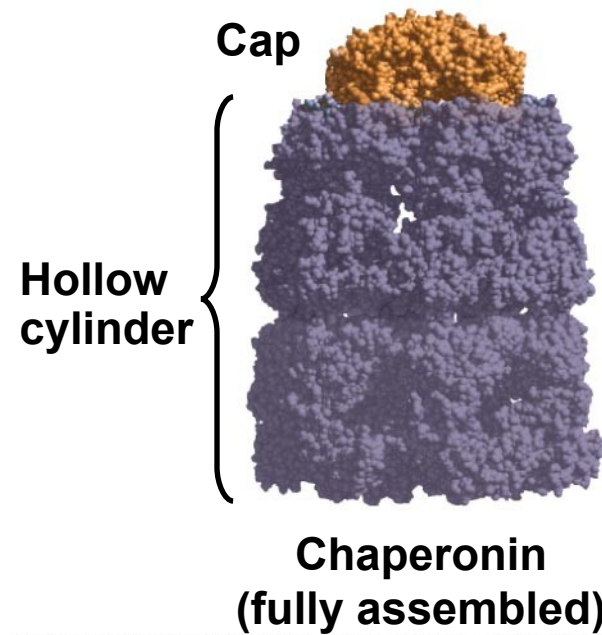
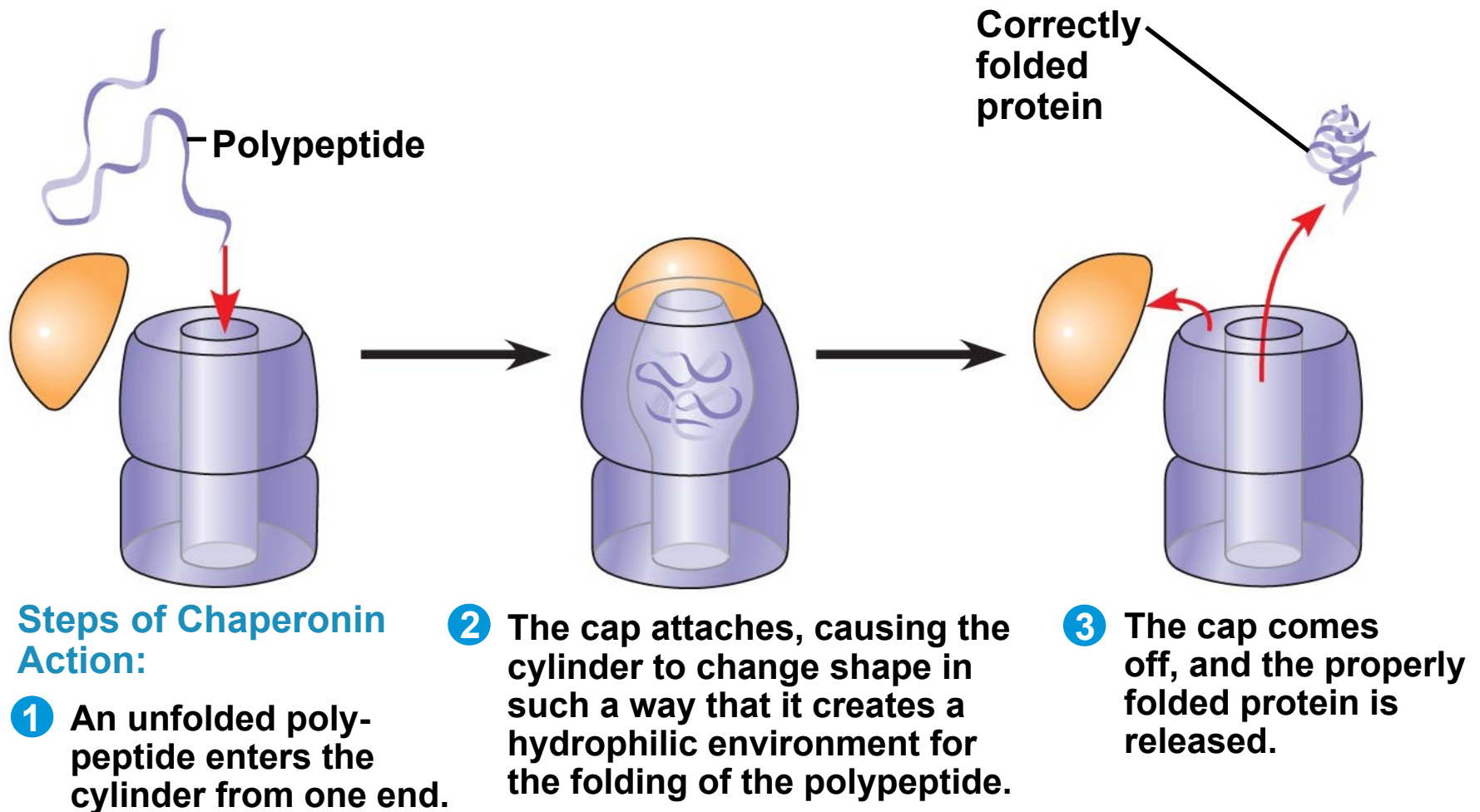


Fig. 5-24a

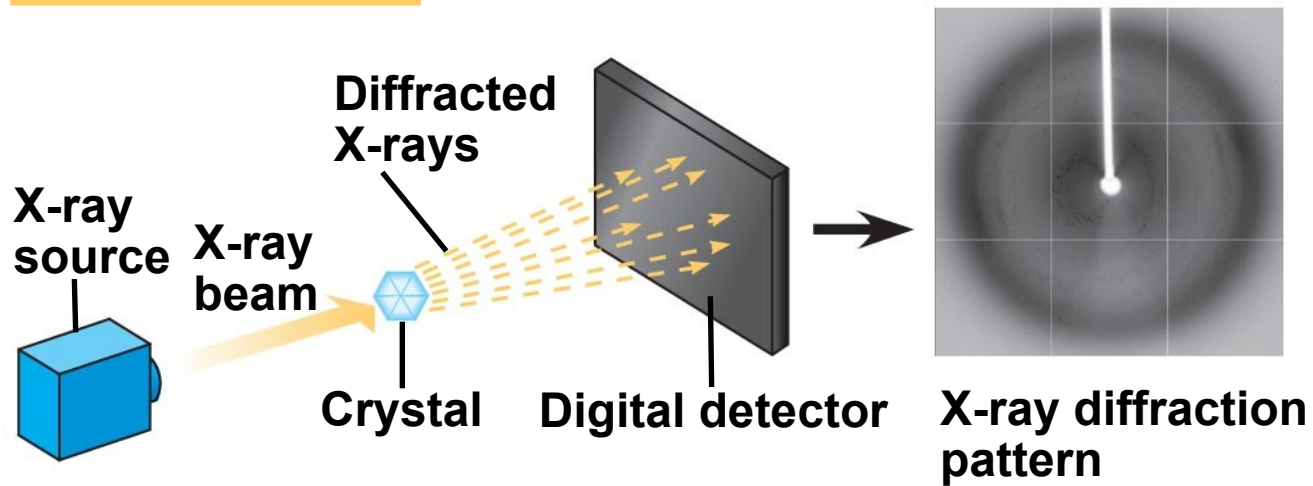


Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

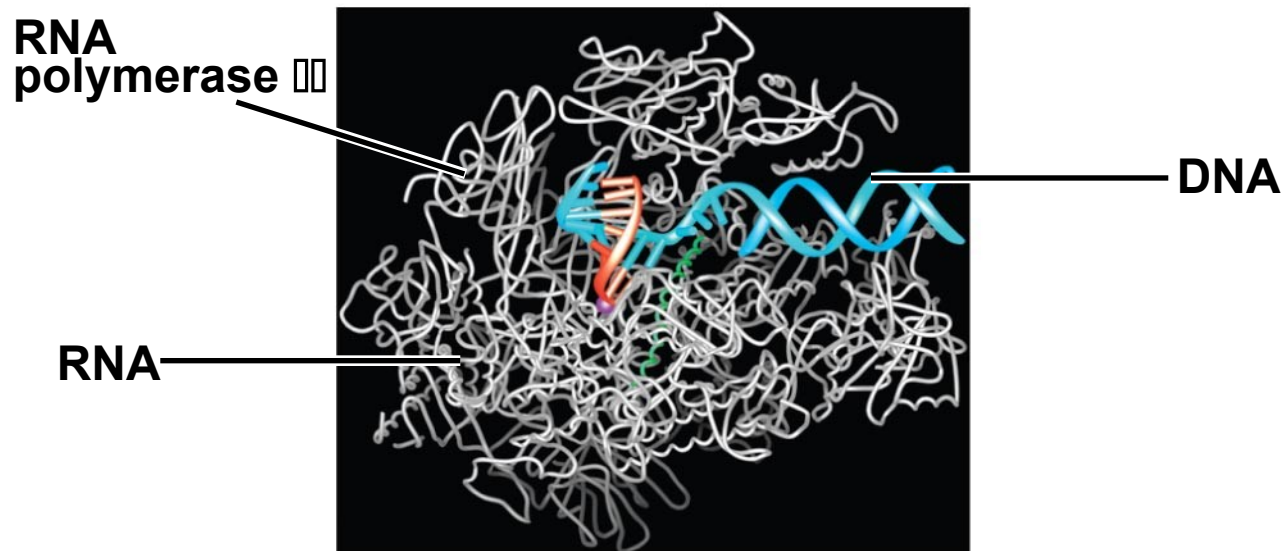


-
- Scientists use **X-ray crystallography** to determine a protein's structure
 - Another method is nuclear magnetic resonance (NMR) spectroscopy, which does not require protein crystallization
 - Bioinformatics uses computer programs to predict protein structure from amino acid sequences

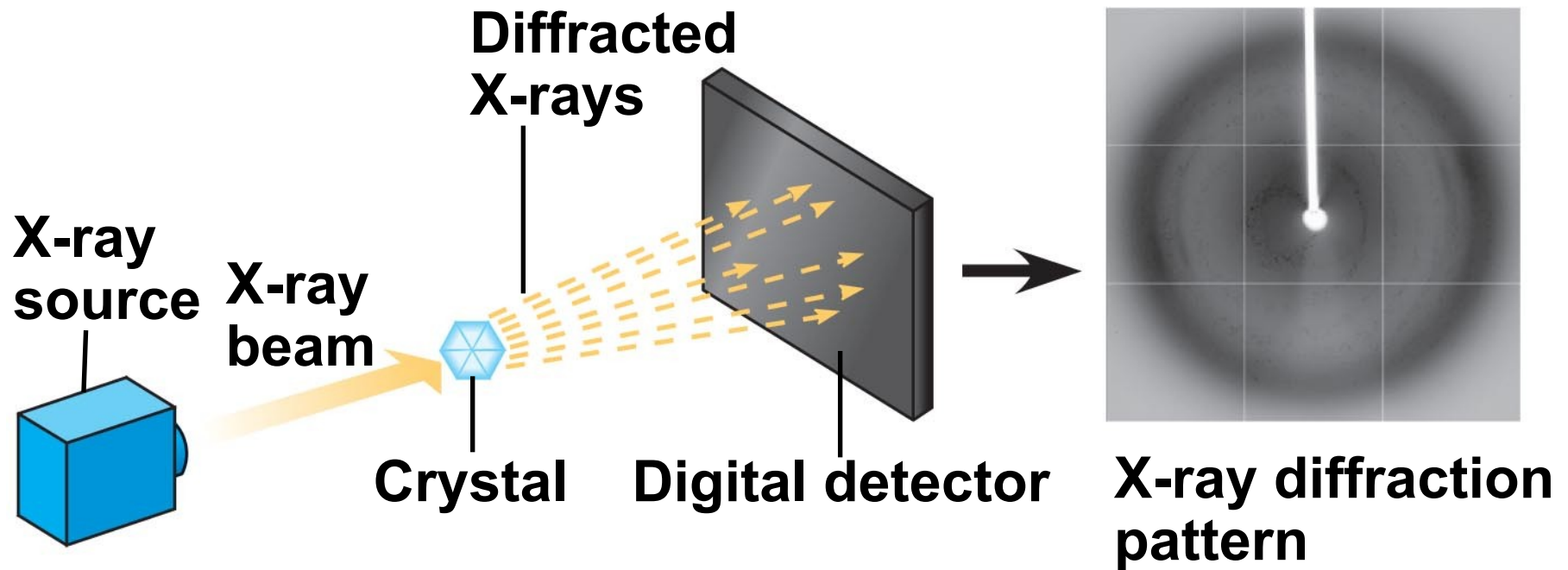
EXPERIMENT



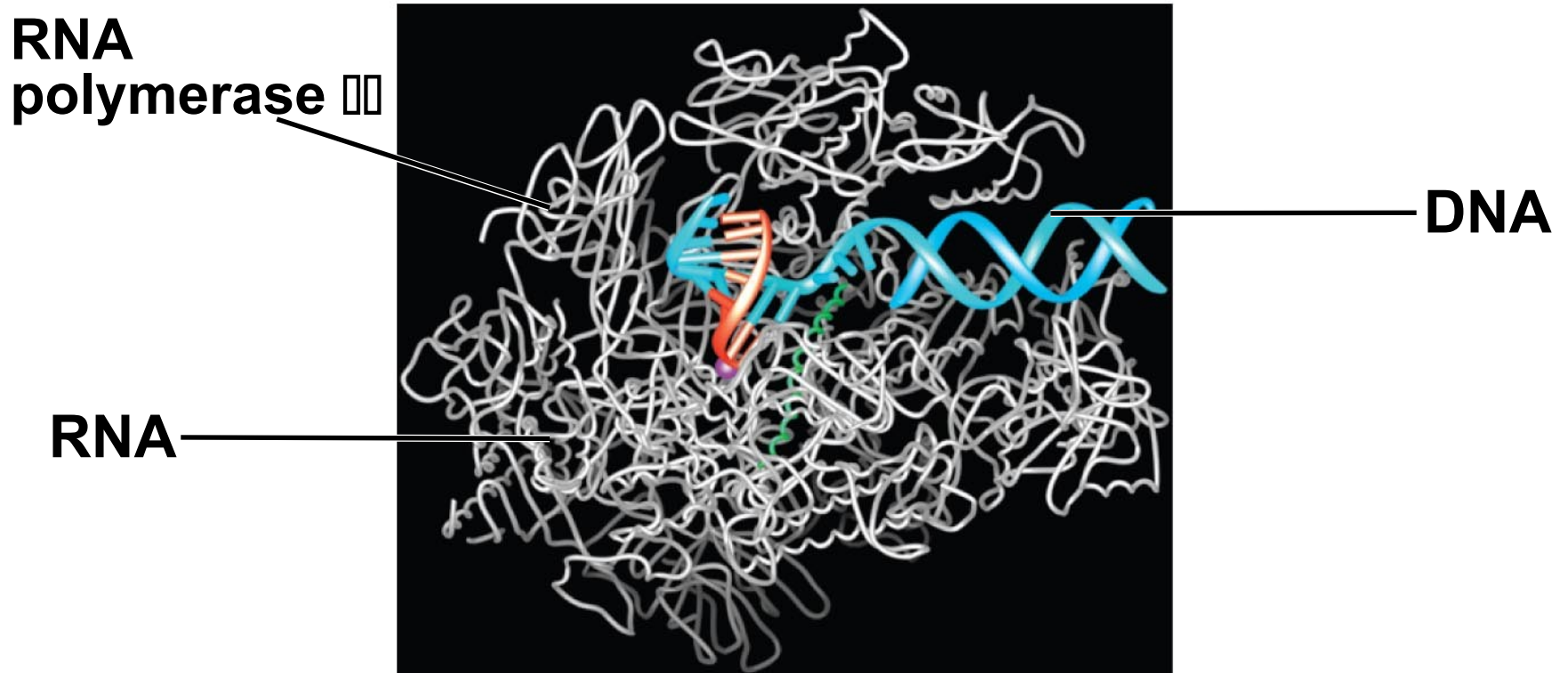
RESULTS



EXPERIMENT



RESULTS



Concept 5.5: Nucleic acids store and transmit hereditary information

- The amino acid sequence of a polypeptide is programmed by a unit of inheritance called a **gene**
- Genes are made of DNA, a **nucleic acid**

The Roles of Nucleic Acids

- There are two types of nucleic acids:
 - **Deoxyribonucleic acid (DNA)**
 - **Ribonucleic acid (RNA)**
- DNA provides directions for its own replication
- DNA directs synthesis of messenger RNA (mRNA) and, through mRNA, controls protein synthesis
- Protein synthesis occurs in ribosomes

Fig. 5-26-1

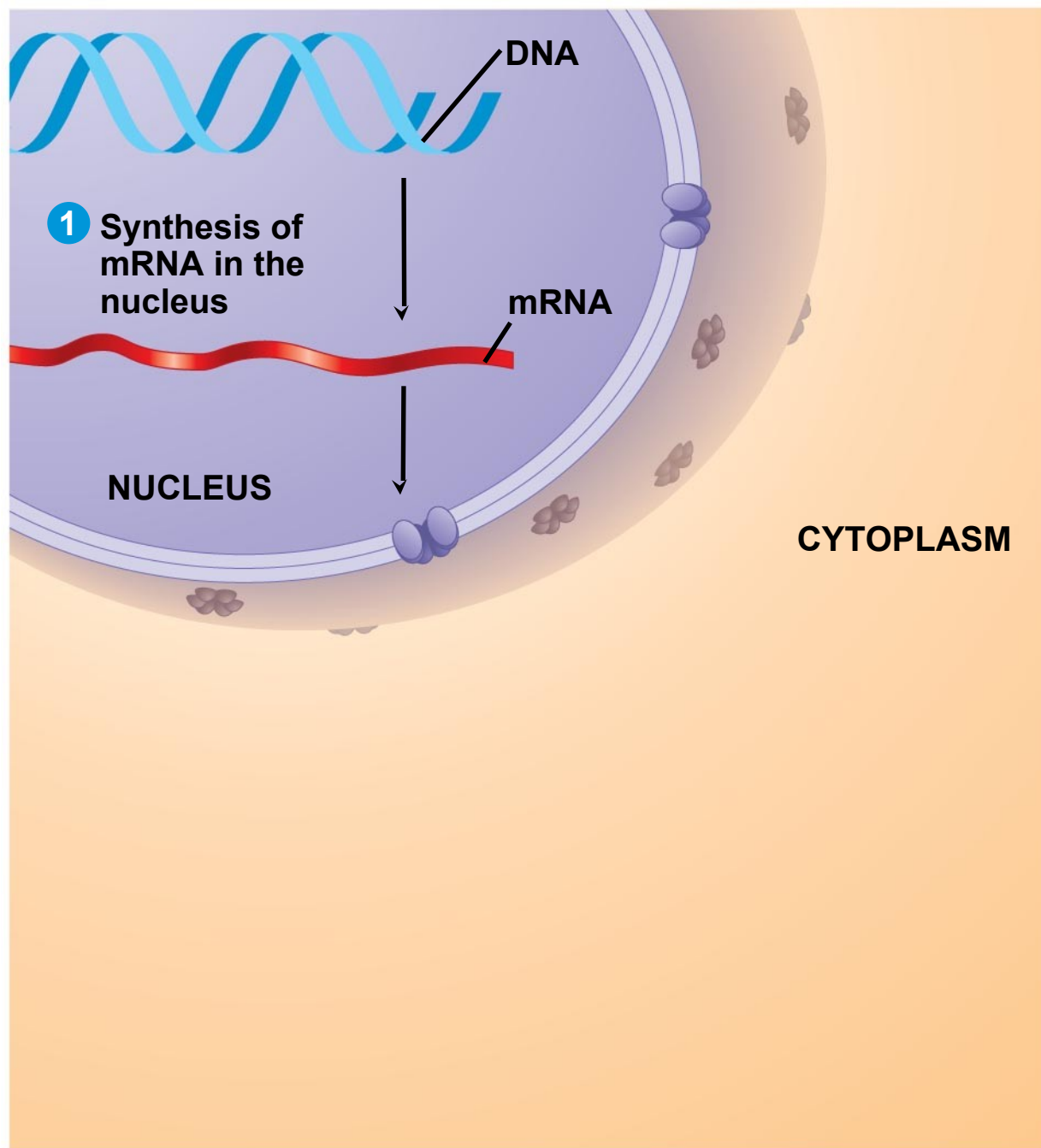


Fig. 5-26-2

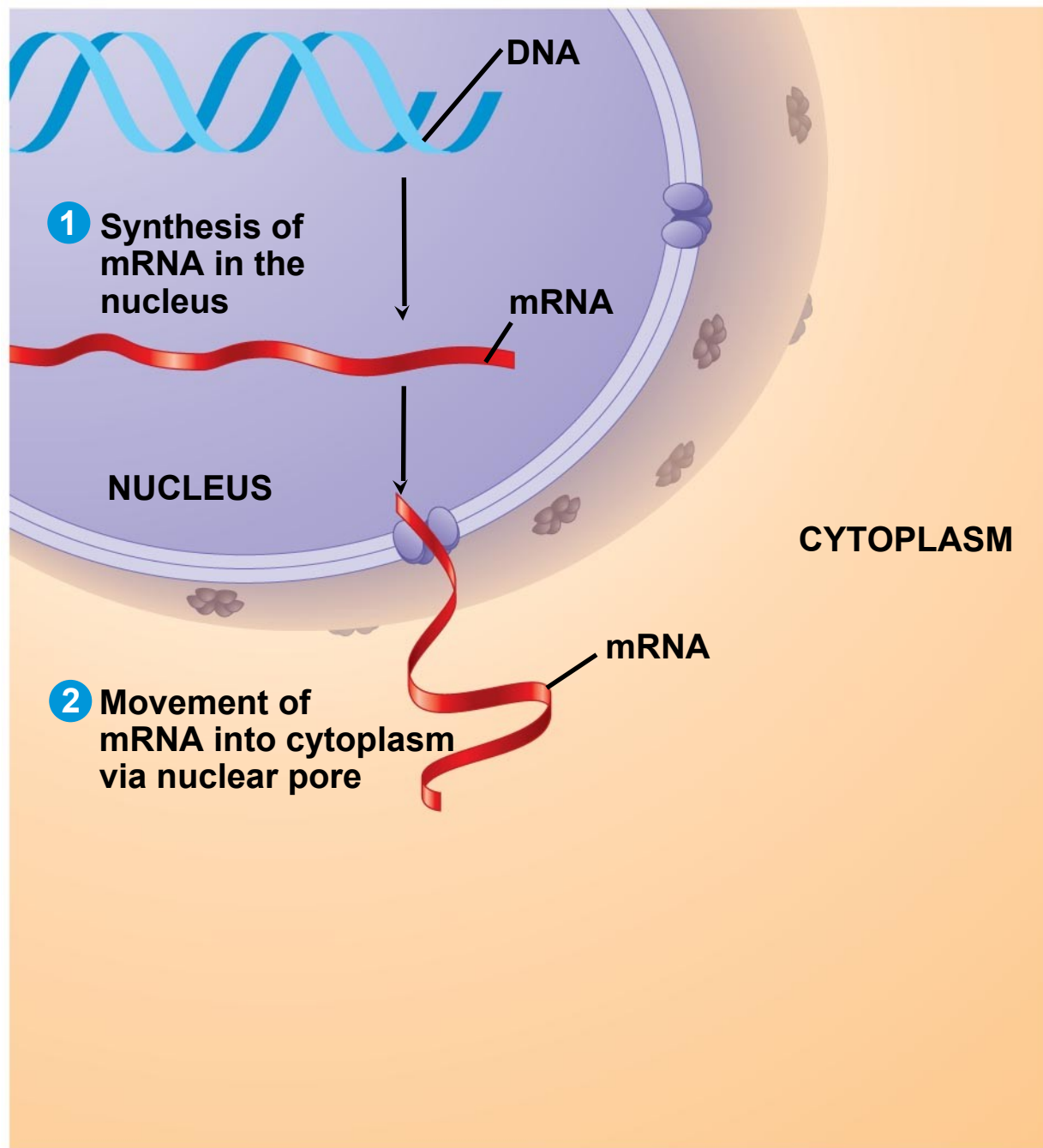
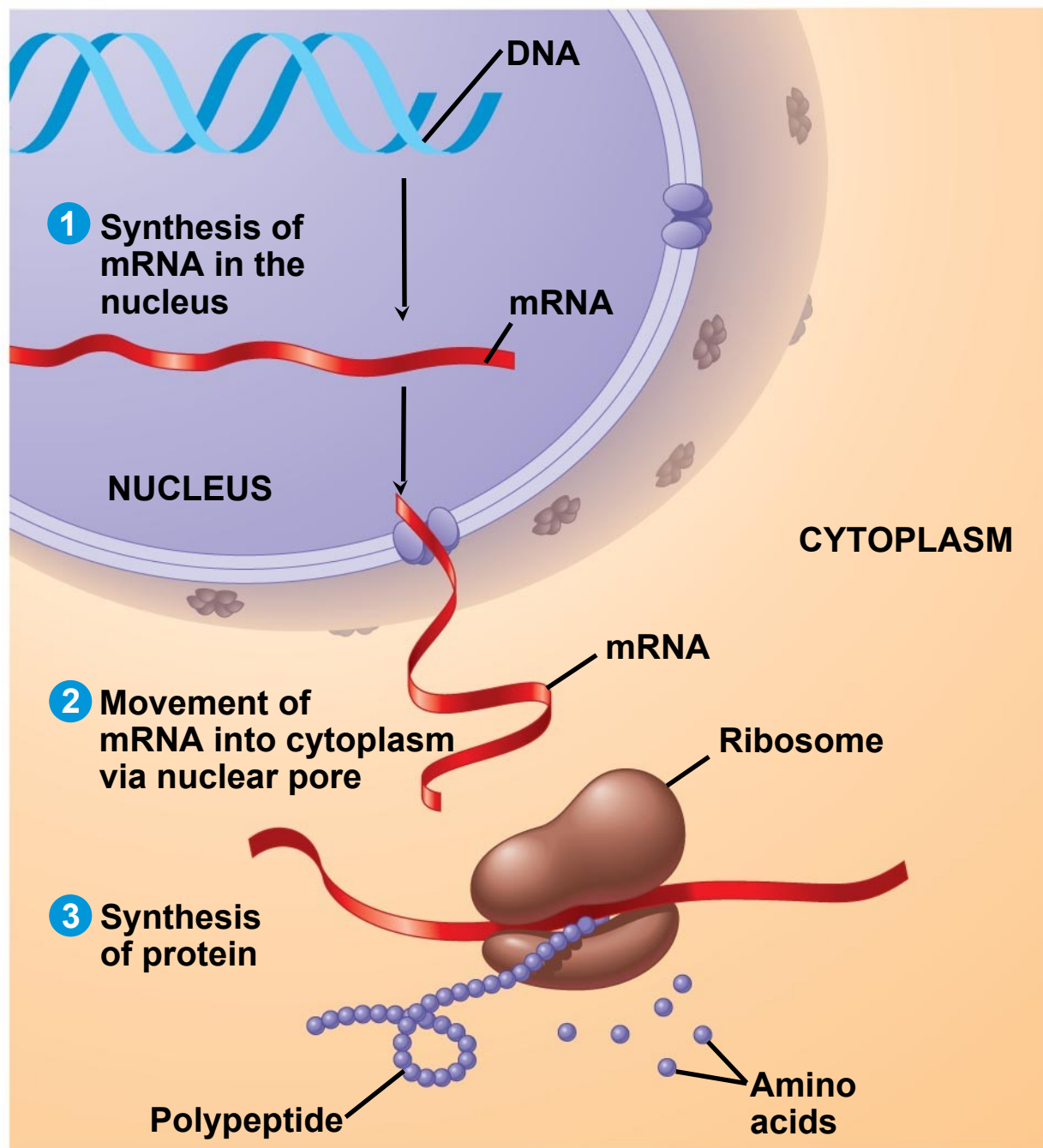


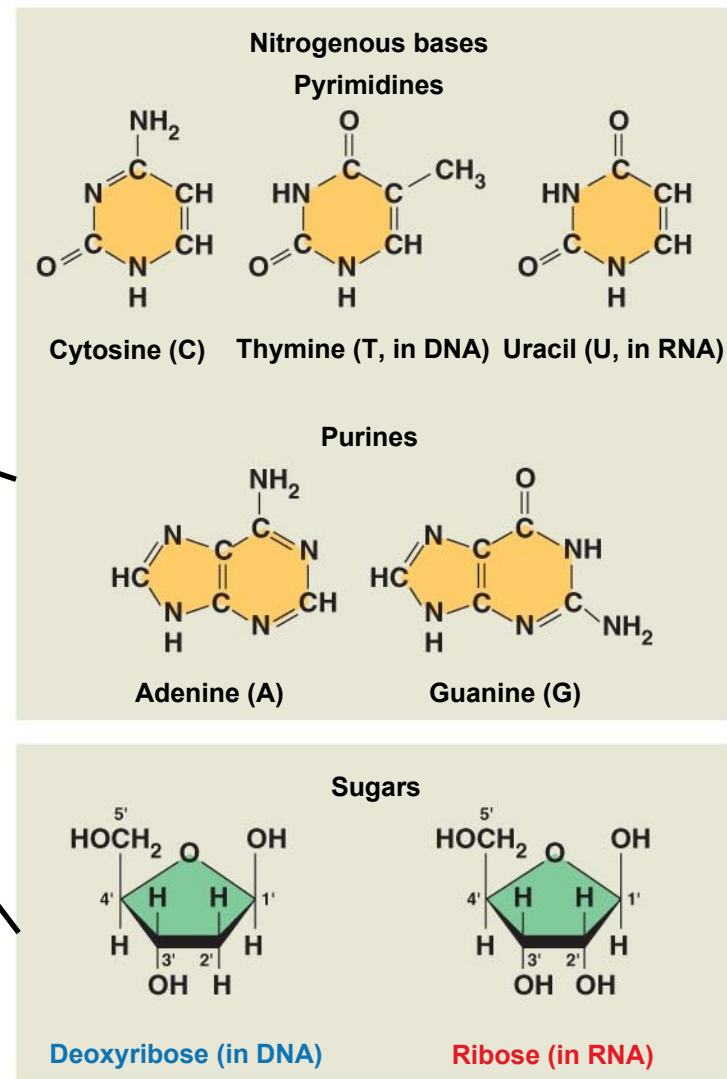
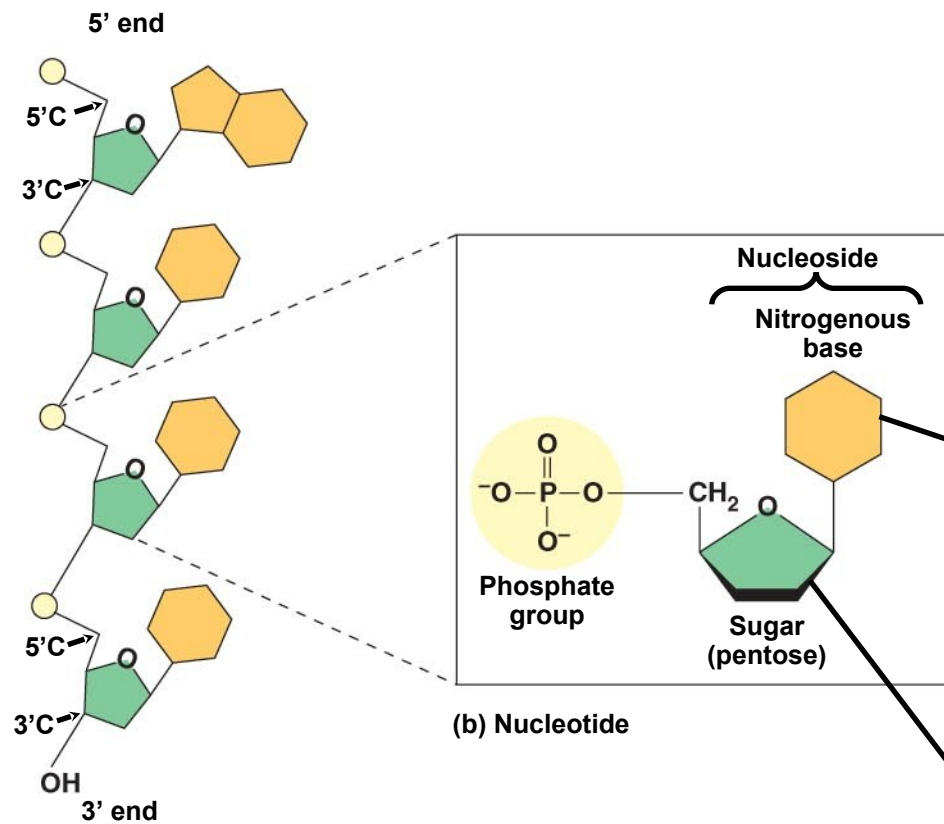
Fig. 5-26-3



The Structure of Nucleic Acids

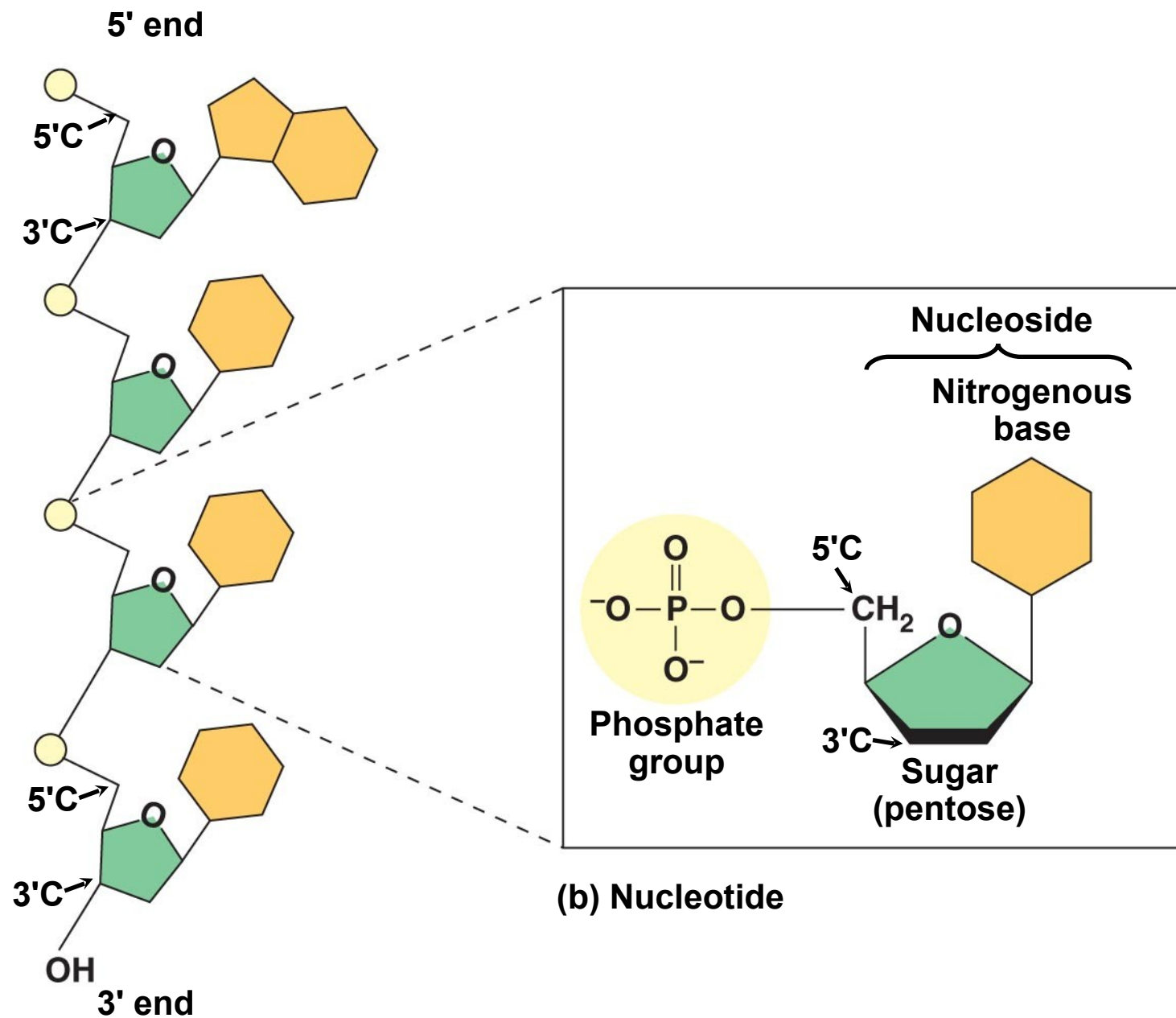
- Nucleic acids are polymers called **polynucleotides**
- Each polynucleotide is made of monomers called **nucleotides**
- Each nucleotide consists of a nitrogenous base, a pentose sugar, and a phosphate group
- The portion of a nucleotide without the phosphate group is called a *nucleoside*

Fig. 5-27



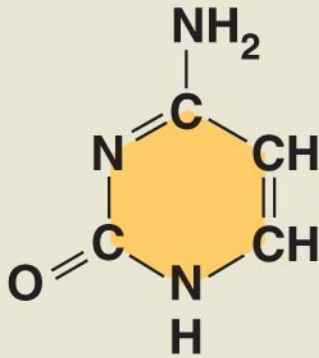
(c) Nucleoside components: sugars

Fig. 5-27ab

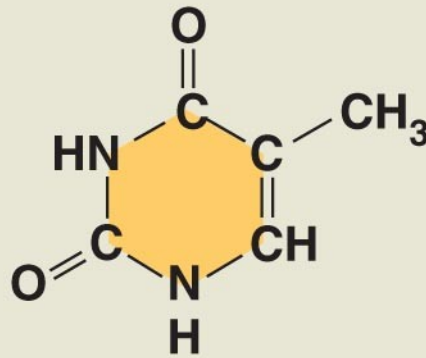


Nitrogenous bases

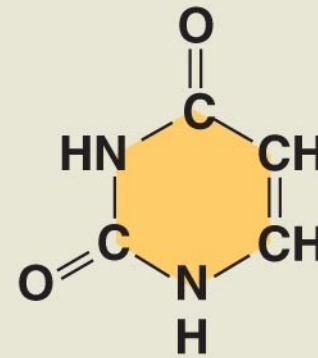
Pyrimidines



Cytosine (C)

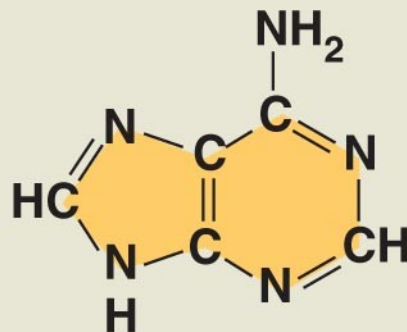


Thymine (T, in DNA)

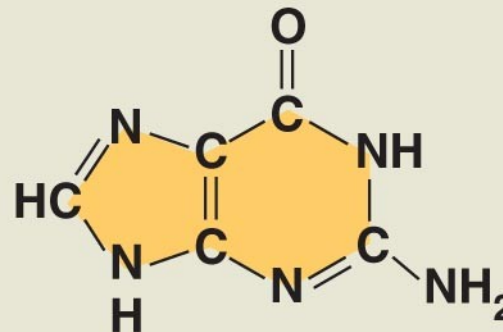


Uracil (U, in RNA)

Purines

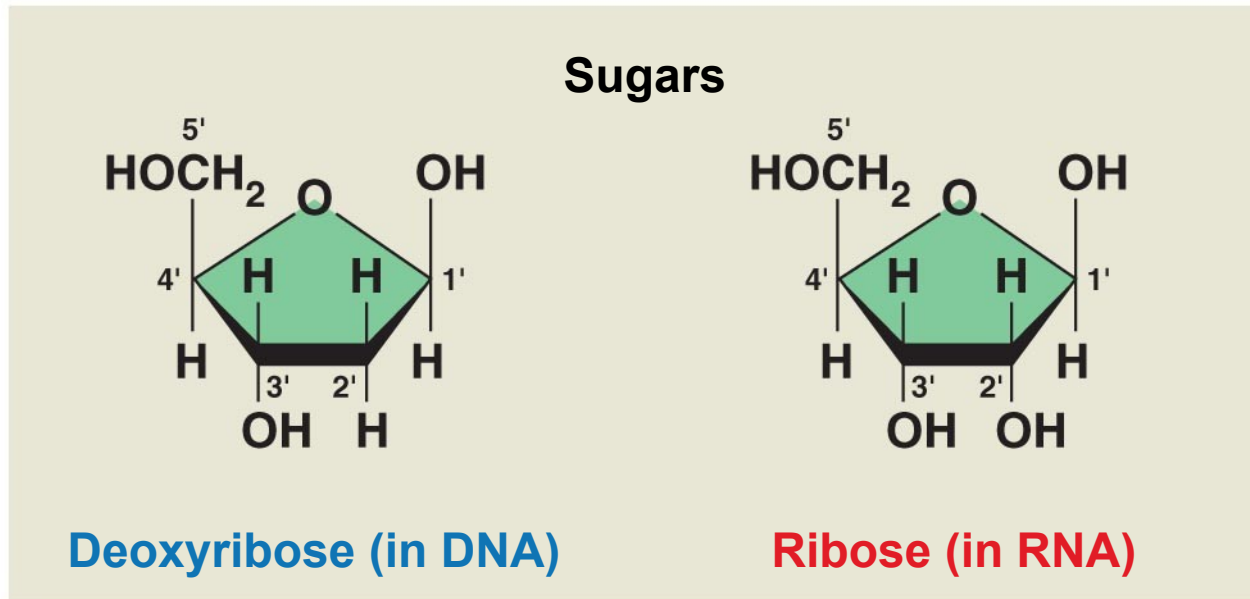


Adenine (A)



Guanine (G)

(c) Nucleoside components: nitrogenous bases



(c) Nucleoside components: sugars

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

Nucleotide Monomers

- Nucleoside = nitrogenous base + sugar
- There are two families of nitrogenous bases:
 - **Pyrimidines** (cytosine, thymine, and uracil) have a single six-membered ring
 - **Purines** (adenine and guanine) have a six-membered ring fused to a five-membered ring
- In DNA, the sugar is **deoxyribose**; in RNA, the sugar is **ribose**
- Nucleotide = nucleoside + phosphate group

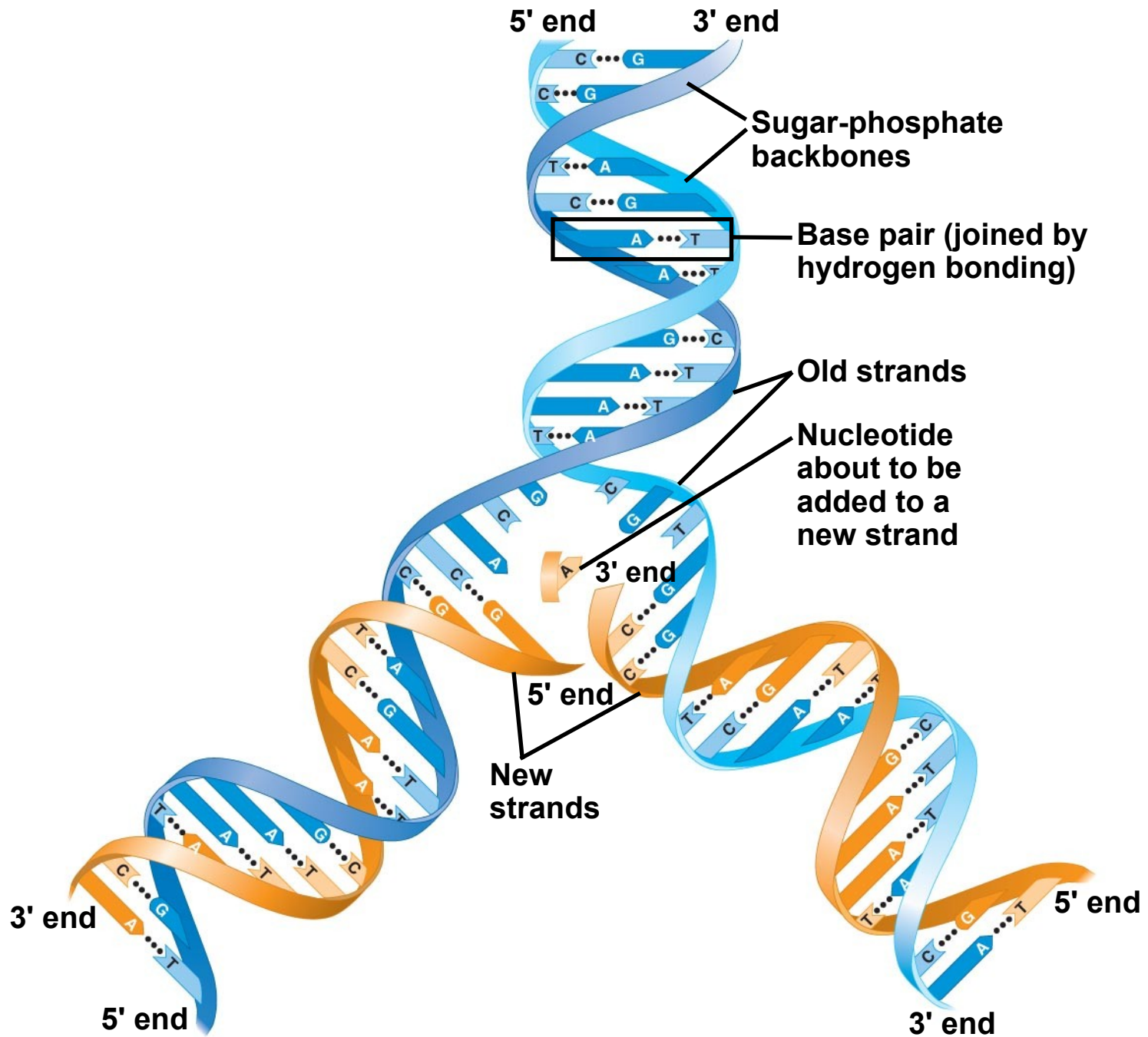
Nucleotide Polymers

- Nucleotide polymers are linked together to build a polynucleotide
- Adjacent nucleotides are joined by covalent bonds that form between the –OH group on the 3' carbon of one nucleotide and the phosphate on the 5' carbon on the next
- These links create a backbone of sugar-phosphate units with nitrogenous bases as appendages
- The sequence of bases along a DNA or mRNA polymer is unique for each gene

The DNA Double Helix

- A DNA molecule has two polynucleotides spiraling around an imaginary axis, forming a **double helix**
- In the DNA double helix, the two backbones run in opposite 5' → 3' directions from each other, an arrangement referred to as **antiparallel**
- One DNA molecule includes many genes
- The nitrogenous bases in DNA pair up and form hydrogen bonds: adenine (A) always with thymine (T), and guanine (G) always with cytosine (C)

Fig. 5-28



DNA and Proteins as Tape Measures of Evolution

- The linear sequences of nucleotides in DNA molecules are passed from parents to offspring
- Two closely related species are more similar in DNA than are more distantly related species
- Molecular biology can be used to assess evolutionary kinship

The Theme of Emergent Properties in the Chemistry of Life: *A Review*

- Higher levels of organization result in the emergence of new properties
- Organization is the key to the chemistry of life

Fig. 5-UN2

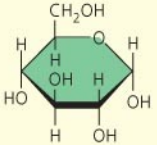
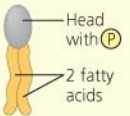
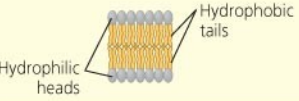
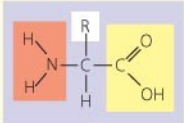
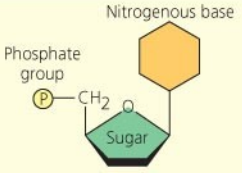
Large Biological Molecules	Components	Examples	Functions
Concept 5.2 Carbohydrates serve as fuel and building material	 Monosaccharide monomer	Monosaccharides: glucose, fructose	Fuel; carbon sources that can be converted to other molecules or combined into polymers
		Disaccharides: lactose, sucrose	
		Polysaccharides: • Cellulose (plants) • Starch (plants) • Glycogen (animals) • Chitin (animals and fungi)	• Strengthens plant cell walls • Stores glucose for energy • Stores glucose for energy • Strengthens exoskeletons and fungal cell walls
Concept 5.3 Lipids are a diverse group of hydrophobic molecules and are not macromolecules	Glycerol  3 fatty acids	Triacylglycerols (fats or oils): glycerol + 3 fatty acids	Important energy source 
	 Head with (P) 2 fatty acids	Phospholipids: phosphate group + 2 fatty acids	Lipid bilayers of membranes  Hydrophobic tails Hydrophilic heads
	 Steroid backbone	Steroids: four fused rings with attached chemical groups	• Component of cell membranes (cholesterol) • Signals that travel through the body (hormones)
Concept 5.4 Proteins have many structures, resulting in a wide range of functions	 Amino acid monomer (20 types)	• Enzymes • Structural proteins • Storage proteins • Transport proteins • Hormones • Receptor proteins • Motor proteins • Defensive proteins	• Catalyze chemical reactions • Provide structural support • Store amino acids • Transport substances • Coordinate organismal responses • Receive signals from outside cell • Function in cell movement • Protect against disease
Concept 5.5 Nucleic acids store and transmit hereditary information	 Nucleotide monomer	DNA:  • Sugar = deoxyribose • Nitrogenous bases = C, G, A, T • Usually double-stranded	Stores all hereditary information
		RNA:  • Sugar = ribose • Nitrogenous bases = C, G, A, U • Usually single-stranded	Carries protein-coding instructions from DNA to protein-synthesizing machinery

Fig. 5-UN2a

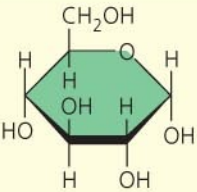
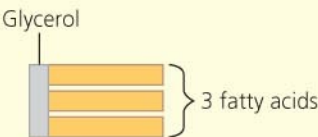
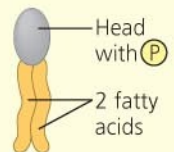
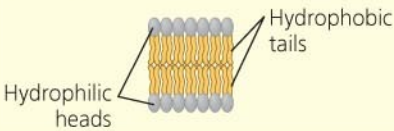
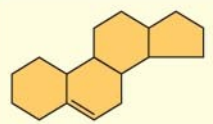
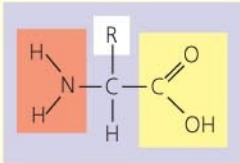
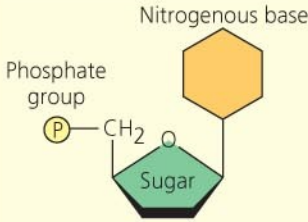
Large Biological Molecules	Components	Examples	Functions
Concept 5.2 Carbohydrates serve as fuel and building material	 Monosaccharide monomer	Monosaccharides: glucose, fructose	Fuel; carbon sources that can be converted to other molecules or combined into polymers
		Disaccharides: lactose, sucrose	
		Polysaccharides: • Cellulose (plants) • Starch (plants) • Glycogen (animals) • Chitin (animals and fungi)	• Strengthens plant cell walls • Stores glucose for energy • Stores glucose for energy • Strengthens exoskeletons and fungal cell walls
Concept 5.3 Lipids are a diverse group of hydrophobic molecules and are not macromolecules	 Glycerol 3 fatty acids	Triacylglycerols (fats or oils): glycerol + 3 fatty acids	Important energy source 
	 Head with P 2 fatty acids	Phospholipids: phosphate group + 2 fatty acids	Lipid bilayers of membranes  Hydrophilic heads Hydrophobic tails
	 Steroid backbone	Steroids: four fused rings with attached chemical groups	• Component of cell membranes (cholesterol) • Signals that travel through the body (hormones)

Fig. 5-UN2b

Large Biological Molecules	Components	Examples	Functions
Concept 5.4 Proteins have many structures, resulting in a wide range of functions	 Amino acid monomer (20 types)	• Enzymes • Structural proteins • Storage proteins • Transport proteins • Hormones • Receptor proteins • Motor proteins • Defensive proteins	• Catalyze chemical reactions • Provide structural support • Store amino acids • Transport substances • Coordinate organismal responses • Receive signals from outside cell • Function in cell movement • Protect against disease
Concept 5.5 Nucleic acids store and transmit hereditary information	 Nucleotide monomer	DNA:  • Sugar = deoxyribose • Nitrogenous bases = C, G, A, T • Usually double-stranded RNA:  • Sugar = ribose • Nitrogenous bases = C, G, A, U • Usually single-stranded	Stores all hereditary information Carries protein-coding instructions from DNA to protein-synthesizing machinery

You should now be able to:

1. List and describe the four major classes of molecules
2. Describe the formation of a glycosidic linkage and distinguish between monosaccharides, disaccharides, and polysaccharides
3. Distinguish between saturated and unsaturated fats and between *cis* and *trans* fat molecules
4. Describe the four levels of protein structure

You should now be able to:

5. Distinguish between the following pairs:
pyrimidine and purine, nucleotide and nucleoside, ribose and deoxyribose, the 5' end and 3' end of a nucleotide